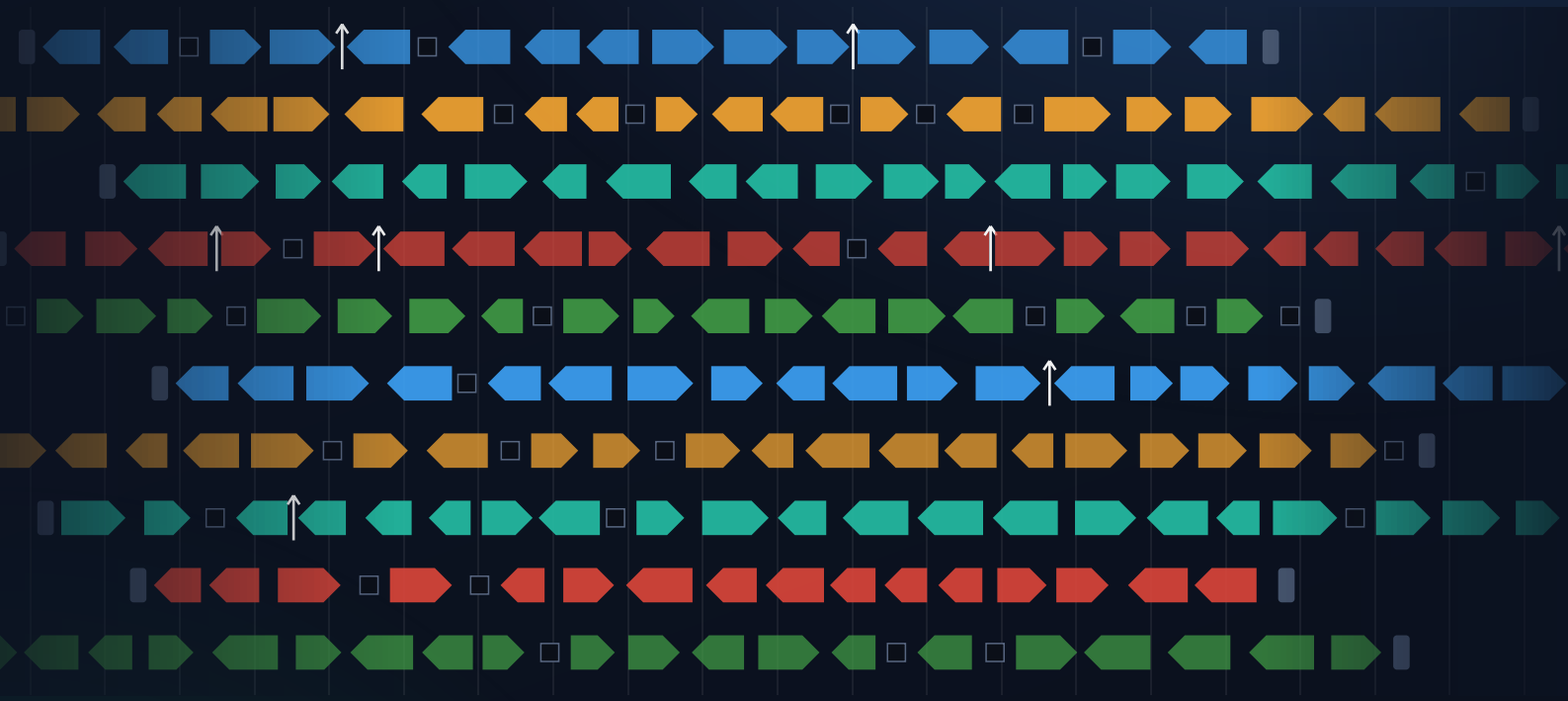
RosterFlow

Bus Scheduling & Crew Rostering · Omsi-Hub

The complete handbook

From an empty line to a day on the road: timetables, blocks, duties and rosters, planned the way a real bus operator does it, explained step by step.



Routes · Timeplan · Vehicles · Drivers · Roster · Optimise · Preferences · Fixing issues

October 2026 edition

For Omsi-Hub 1.1 with the RosterFlow add-on

Manual · course book · reference

About this handbook

RosterFlow is the professional scheduling studio of the bus company in Omsi-Hub. This handbook teaches you how it works, why it works that way, and how to use it to plan a good, cheap and reliable day.

Who this book is for

For anyone who plays a company in the **Normal** or **Realistic** mode and wants to take the planning into their own hands. You need no experience with real scheduling software: every term is explained before it is used. If you are already experienced, part IV lets you find everything quickly: every button, every rule, every issue and every shortcut.

How to read it

PART I · UNDERSTAND

What RosterFlow does

The chain of planning steps, the words schedulers use and how to read the screen. Read this first, even if you are in a hurry.

PART II · PLAN

Every step in practice

Routes, timetable, vehicles, drivers, roster and saving, in the order you really do them.

PART III · REFINE

Better and cheaper

Optimising, fixing issues, the rules in Preferences, copying to other days, printing and planning together.

PART IV · APPLY

Recipes, exercises and reference

Complete workflows for common situations, exercises with checkpoints and every table to look things up.

Signs in the margin

TIP
A trick that saves time or money.

WATCH OUT
Something players often trip over.

HOW IT WORKS
What RosterFlow does behind the scenes. Handy to understand why something happens.

TRY IT YOURSELF
A short task to do right away. You can undo everything.

Conventions

Buttons and tabs are written in **bold**, as they appear on screen; capitalised buttons such as SAVE are shown in capitals too. Keys look like Ctrl + Z, and a path through menus like **Preferences › Drivers › Duty Breaks**.

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01

PART I

Understand

Before you move a single trip: what RosterFlow plans, in what order, with which words, and how to read the screen.

1 What RosterFlow is

2 The scheduler's language

3 Reading the screen

CHAPTER 1

What RosterFlow is

RosterFlow plans your company's day: when the buses run, which bus drives which trips, and which driver drives which bus. It is a chain of steps, and nothing counts until you save.

Why a scheduling studio?

Without RosterFlow the game largely handles the planning for you: a line gets its tours, a tour gets its duties, and you assign drivers and buses. That is quick, but you have little say in *how* the day is put together. With RosterFlow you decide everything yourself, exactly as a scheduler at a real operator does with professional software:

- how long a bus takes over each stretch of route, and how often it runs, for each part of the day;
- which bus drives which trips in a row, with as few empty kilometres as possible;
- where and when a driver takes over the bus, when they take a break and how long their working day is;
- which driver and which bus run on which day of the week;
- which rules all of that must keep, and what it costs when a rule is broken anyway.

The result is a planning the game then really runs: the roster, the company day and the buses on the map follow what you saved in RosterFlow.

Switching RosterFlow on and opening it

RosterFlow is an **add-on** of the bus company. You switch it on in the company's settings, where you also choose the game mode. It exists in the **Normal** and **Realistic** modes; in **Casual** the game plans everything itself and RosterFlow is off.

1. **Open your company**
   In Omsi-Hub, go to the bus company and open the company you want to plan.
2. **Switch the add-on on**
   Switch RosterFlow on in the company's settings. You see the message *RosterFlow is on: the planning is its studio now.*
3. **Open the Planning page**
   Choose **Planning** in the menu on the left. Instead of the usual day roster, RosterFlow now opens, with its own title bar and five tabs.
4. **Back to the company**
   The arrow on the left of the title bar, or Esc, takes you back to the company's start screen.

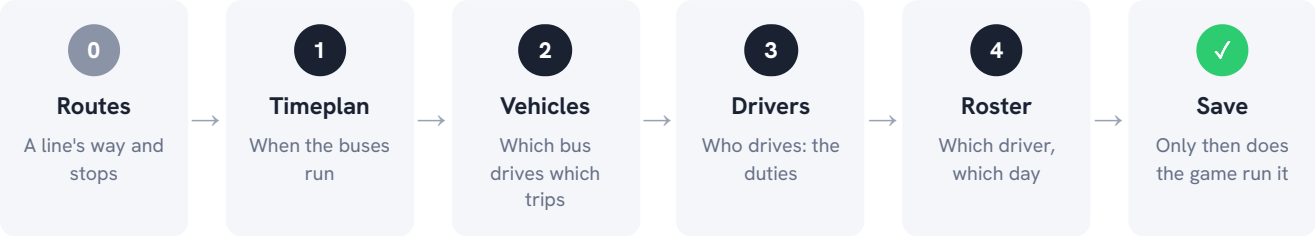
> **HOW IT WORKS**
> If you switch RosterFlow off later (or move to Casual), all its plans are kept. The planning is then the ordinary day roster again. Switch it on again and everything is still there.

TIP

If your company is in the cloud, RosterFlow also works in the browser on omsihub.com. It is the same studio; only printing to a PDF file is for the computer.

The chain: five tabs, one direction

Every planning in RosterFlow follows the same chain. Each step builds on the one before, and the tabs at the top are in that order:



0 · ROUTES

The line itself

A line gets its route over the roads and its stops, with the minutes from stop to stop. You do this once per line (or when the route changes).

1 · TIMEPLAN

When the buses run

The running times between stops, per time band, and the departure times of every trip. Every other step is built from these trips.

2 · VEHICLES

Which bus drives which trips

The day's trips are shared out among your buses. One bus's day of trips is called a **block**. A good block keeps the bus busy and has few empty runs.

3 · DRIVERS

Who drives

A bus's day is too long for one driver, so each block is cut into **duties**: one driver's working day. A cut can only be made where a driver can get on or off.

4 · ROSTER

Which driver, which day

Your drivers get their duties and your buses their blocks, weekday by weekday. What you plan for a Monday repeats every Monday.

✓ · SAVE

Draft becomes planning

Your changes in Vehicles and Drivers are a **draft**. SAVE makes them the company's planning; DISCARD throws the draft away.

WATCH OUT: THE CHAIN'S DIRECTION

Changes always flow *to the right*. Change the timetable, and the blocks and duties no longer match the new trips: RosterFlow says so with **UNSYNCED**, and you choose **Refresh** to make them anew (chapter 5). The other way round, cutting a duty changes nothing in the timetable.

The day you plan: weekdays, not dates

RosterFlow always plans **one day** at a time, and you choose that day at the top right of the title bar. The most important thing to remember: a day's planning holds **for that weekday every week**. Change a Tuesday, and every Tuesday changes. That way you only plan a normal week once.

If you do want one particular date to be different (a holiday, an event), you can choose **This date only** when saving. That exception holds for that one day; afterwards the weekday runs its usual planning again. Chapter 10 explains exactly how.

Draft and save: nothing counts until you save

Everything you do in **Vehicles** and **Drivers** is a draft at first. The game keeps running the saved planning meanwhile. You can tell the state from the dot beside your company's name in the title bar:

DOT	MEANING
Green	Saved: the company runs exactly what you see.
Orange	There are changes not saved yet. SAVE makes them the company's planning, DISCARD forgets them.

That leaves you free to experiment: move, cut, optimise and compare. Don't like it? Discard everything in one click, or undo step by step with Ctrl + Z.

> **WATCH OUT**
> Don't forget to save before you let the day run. If RosterFlow has unsaved changes while the day runs, the day simply runs the *saved* blocks and duties, and the company says *RosterFlow not saved*.

Help inside the studio

RosterFlow has its own help behind the question mark in the title bar: a guided tour of every part of the screen, short lessons in which you do a task yourself, and the help sheet *RosterFlow in 2 minutes*. This handbook goes much further, but the lessons are a fine way to practise. Chapter 16 covers them.

CHAPTER 2

The scheduler's language

Bus scheduling has its own words. Know them, and you understand every screen of RosterFlow straight away. Take your time with this chapter; you will meet every word again in the rest of the book.

One bus, one day: a block in pictures

The diagram below shows the main terms in one picture. At the top, one bus runs all day: its **block**. Below it, that block is cut into two **duties**: the working days of two drivers, who **relieve** each other at a terminus.

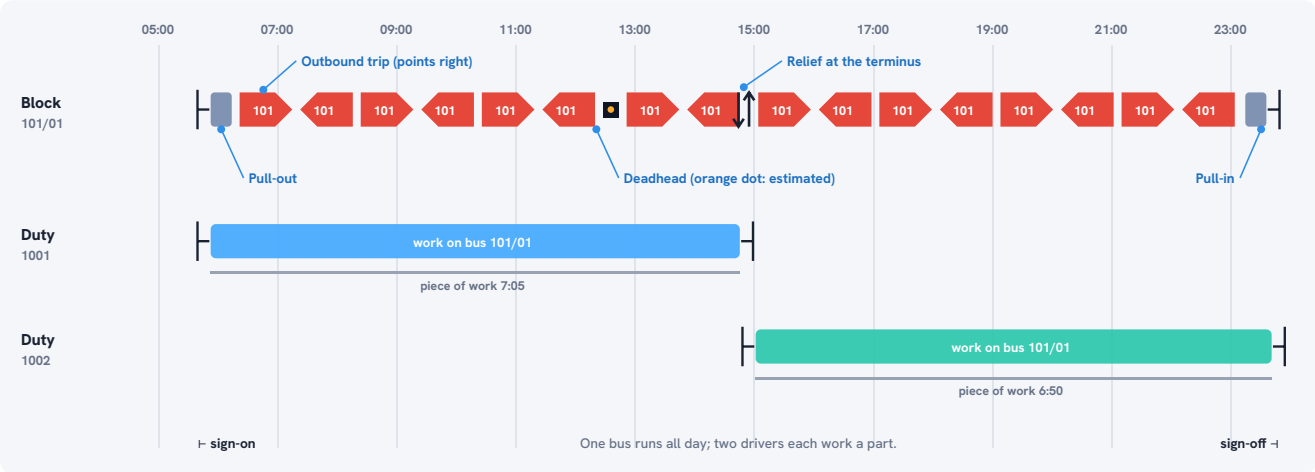


Figure 2.1 A block (top) is cut at a terminus into two duties (below). The arrows mark the relief; the T-shaped marks are the sign-on and sign-off.

Every term at a glance

Trip

One run of a bus from one end of the line to the other, with passengers. Shown as a coloured **chip** in the line's colour.

Direction outbound / inbound

The pointed end of a chip shows the way it runs: right is outbound, left is inbound.

Block vehicle schedule; in the Roster: tour

One bus's whole day: every trip it drives, from leaving the depot until it is back. Named like **101/01**: line 101, block 01.

Duty

One driver's working day, from sign-on to sign-off. Duties have numbers such as **1001**.

Piece of work

The part of a duty on one bus. A duty can have pieces on different buses.

Relief

Where one driver hands the bus to the next: always at a **relief point**, such as a terminus or the depot, or a stop on the way that you set aside for it.

Relief point

A place where duties may begin or end, drivers may change and take breaks. By themselves: the depot and the lines' termini.

Sign-on and sign-off

The start and the end of a duty, with a few paid minutes to check the bus and close the duty.

Deadhead

A bus running without passengers, for example to where its next trip starts. An orange dot means its time is a guess.

Pull-out and pull-in

The bus leaving the depot in the morning and going back in the evening. Grey-blue boxes at the start and end of a block.

Layover

The wait at the end of the line before the next trip. Too short is unreliable, too long is wasted bus time.

Break and meal break

A gap in a duty in which the driver doesn't drive. A duty that works long needs a **meal break** of a minimum length.

Split duty split

A long unpaid break in the middle of a duty: the driver works early and again later. On screen a line with the word *Split*.

Duty travel

How a driver gets to the next bus: **W** walking, **P** by public transport, **R** by road in a staff car.

Spread

The time from sign-on to sign-off: how long the working day lasts from start to end, breaks included.

Paid, work, driving and platform time

Paid is what the driver earns; driving time is behind the wheel with the bus moving; platform time is the time on the bus (driving plus layovers).

Stack unassigned

Trips no bus drives yet. They wait above the rows until you give them a bus.

Drop slot

Where something picked up may go, with stars for how good it is: blue breaks no rule, red breaks one.

Locked

A locked bus or duty stays as it is: Optimise and your other changes leave it alone.

Draft

Your changes before SAVE (orange dot). The game keeps running the saved planning (green dot).

PVR peak vehicle requirement

The most buses out at the same time. The day needs at least this many buses.

Time band

A part of the day with its own running times: the rush hour is often slower than the evening.

Running time

The minutes a bus needs from one stop to the next.

Headway

How many minutes apart the buses leave: a 10-minute headway is six buses an hour.

Pattern

One version of a line's route: the whole run, or a short one that skips stops.

Timepoint

A stop whose time counts and is shown in the timetable. Other stops are plain stops.

Day type day group

The group of days that run the same timetable: Monday to Friday, Saturday, Sunday.

Interlining

A bus running trips of several lines in its day.

Spare duty and custom work

A duty without trips, or time in a duty that isn't driving: spare, office, union or training.

Issue

A broken rule, such as a duty without a meal break. Red is a **hard** rule, orange a **soft** one that costs a penalty.

Symbols on the screen

The charts in Vehicles and Drivers use the same symbols as professional scheduling software. Learn them once and you read every row at a glance.

SYMBOL	MEANING
Coloured chip	A trip with passengers; the point is its direction
Grey-blue box	Pull-out from or pull-in to the depot
Small dark box	Deadhead; with an orange dot: time estimated
T-mark ⊢ ⊣	A duty's sign-on and sign-off
Arrows ↓↑	Change of driver (relief)
Line with *Split*	Split duty: long unpaid break
Amber box W, P, R	Driver on the way to the next bus
Green pill 0:35	A break, with its length
White box	Custom work in a duty
Striped chips	Linked trips: they stay together
Red "!"	This row breaks a rule

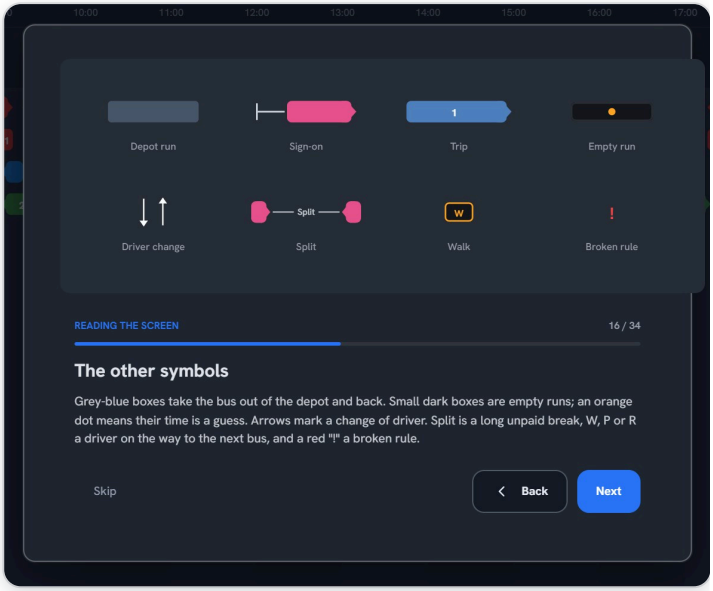


Figure 2.2 The symbols as RosterFlow's guided tour explains them.

CHAPTER 3

Reading the screen

A walk past everything you see when RosterFlow is open: the title bar, the KPIs, the toolbar, the chart and the status bar.

> **ABOUT THE PICTURES**
> The screenshots come from the sample company **Grundorf Linien**, with three lines (101, 102 and 205), four buses and seven duties. Your company looks different, but every screen works the same way. A few labels in the sample data (such as the Roster's line headings) are the sample company's own.

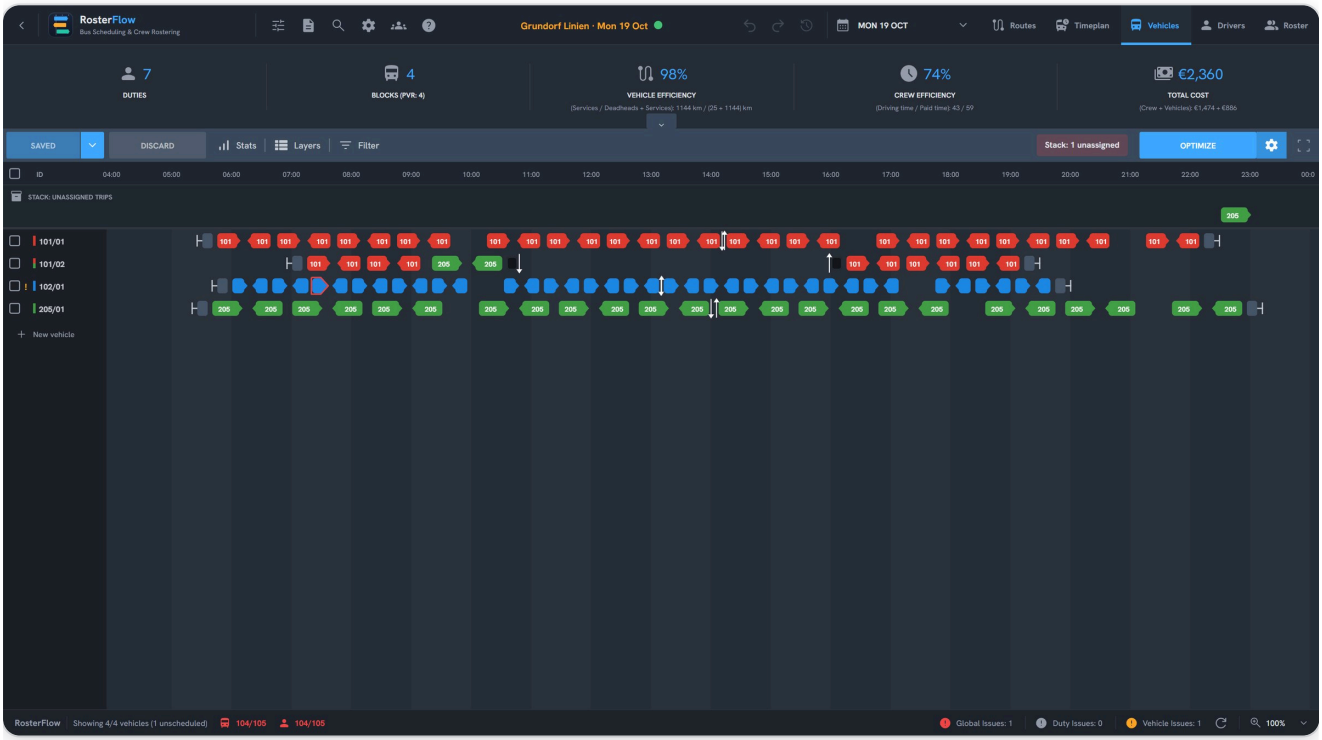


Figure 3.1 RosterFlow on the Vehicles tab. From top to bottom: title bar, KPIs, toolbar, the stack, the rows of blocks and the status bar at the foot.

The title bar

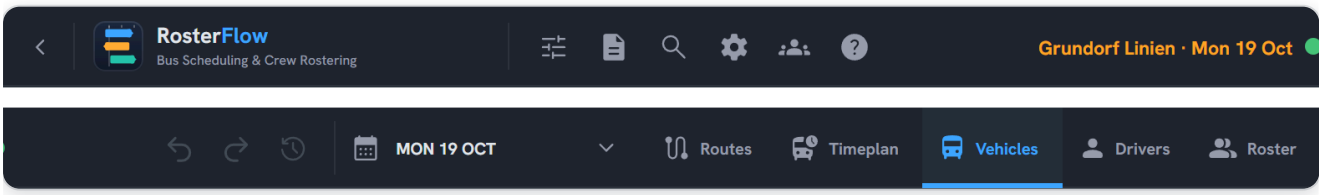


Figure 3.2 The title bar: back, the brand, the icons, the name and the day with its dot, undo and history, the day switcher and the five tabs.

PART	WHAT IT DOES
Left arrow	Back to the company's start screen (Esc).

PART	WHAT IT DOES
Sliders	**Preferences**: the rules the duties are checked against and Optimise works to (chapter 12).
Document	**Print**: the bus graph or the duty graph as a PDF file (chapter 14).
Magnifier	**Go to** a vehicle or a duty by its ID (G).
Gear	**Optimisation settings**: what Optimise makes (chapter 9).
Group of people	**Play together**: friends drive duties of your company in your world (chapter 15).
Question mark	The **tutorial**: guided tour, lessons and the help sheet (chapter 16).
Company · day + dot	Your company's name and the day shown. The dot is 🟢 green when everything is saved and 🟠 orange with unsaved changes.
Arrows and clock	**Undo** (Ctrl + Z), **redo** (Ctrl + Y) and the **schedule history**: every change, to go back to.
Calendar	The **day switcher**: the day you plan, from the coming week.
Tabs	**Routes**, **Timeplan**, **Vehicles**, **Drivers** and **Roster**.

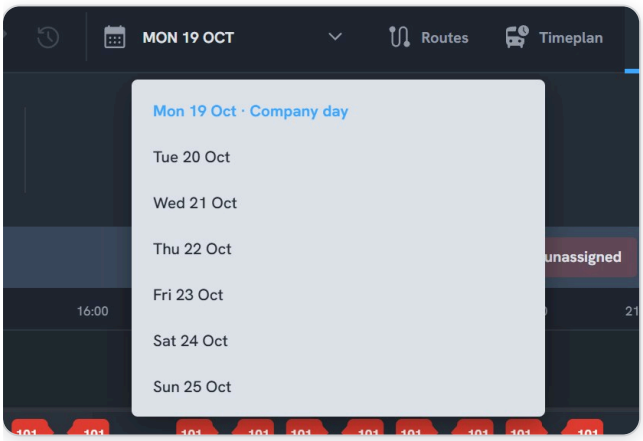


Figure 3.3 The day switcher: the company day (today in the game) and the six days after it.

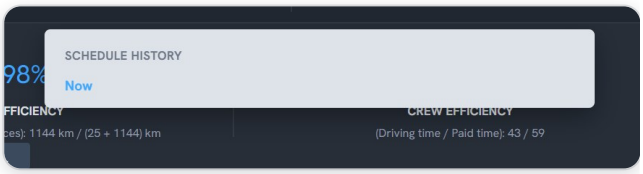


Figure 3.4 The schedule history. Click a line to go back to before that change.

The KPIs

Under the title bar are five figures that together tell how good your day is. They are there in Vehicles and Drivers, and they change live with every edit.

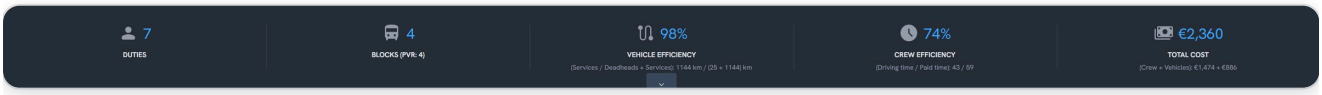


Figure 3.5 The five KPIs. The tab underneath folds out the extended figures.

KPI	WHAT IT MEASURES	BETTER IS
Duties	How many drivers' working days the day needs.	Fewer, as long as every trip is driven
Blocks (PVR)	How many buses run. In brackets the **peak**: how many are out at the busiest moment.	Fewer
Vehicle efficiency	The share of the buses' kilometres that carry passengers: service km divided by deadhead + service.	Higher
Crew efficiency	The share of paid time spent driving: driving time divided by paid time.	Higher
Total cost	What the day costs: crew plus vehicles, at the rates in Preferences › Costs.	Lower

After every change a small difference appears beside a figure: **green** is better, **red** is worse. You see straight away what a move gains or costs.

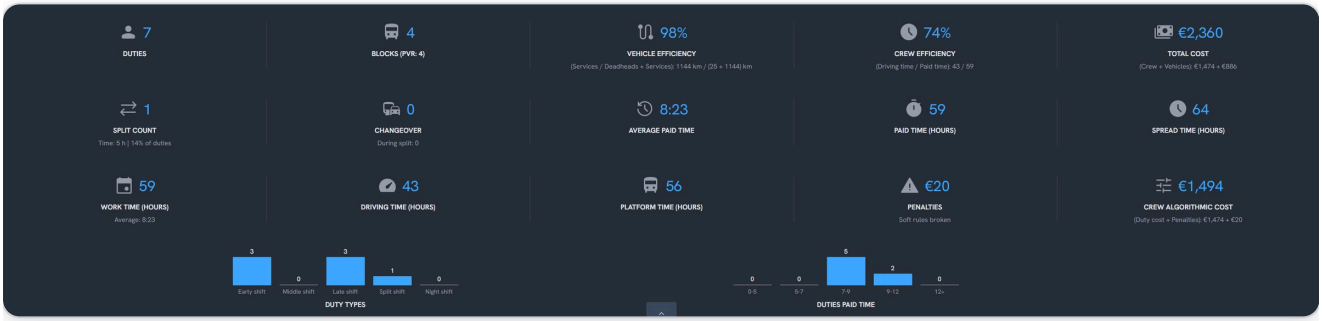


Figure 3.6 The extended figures: splits, changeovers, paid, spread, work, driving and platform time, penalties, the algorithmic cost and two histograms.

EXTENDED FIGURE	EXPLANATION
Split count	How many duties have a split, with the total time and the share of duties.
Changeover	How often drivers change buses, and how often during a split.
Average paid time	The average paid length of a duty.
Paid, spread, work, driving and platform time	The total in hours over all the day's duties.
Penalties	What the broken soft rules cost.
Crew algorithmic cost	Duty cost plus penalties: the number Optimise tries to get as low as possible.
Duty types	Histogram: how many early, middle, late, split and night shifts.
Duties paid time	Histogram: how many duties are paid 0–5, 5–7, 7–9, 9–12 and 12+ hours. Click a histogram to open it large.

The toolbar

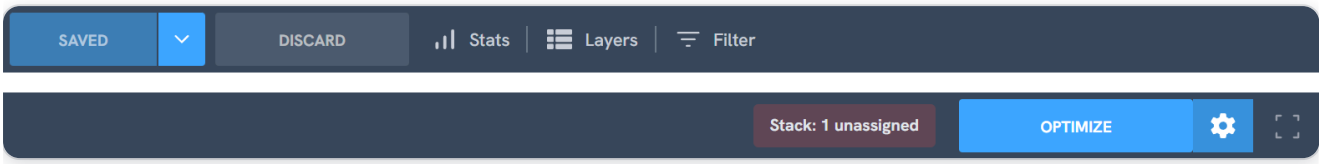


Figure 3.7 The toolbar above the chart.

BUTTON	WHAT IT DOES
SAVE **and arrow**	Makes the draft the company's planning. The arrow opens **Save and apply to other days**. When everything is saved, it reads *Saved*.
DISCARD	Forgets the draft and goes back to the saved planning.
Stats	Adds columns of figures beside the rows. Click a column heading to sort.
Layers	Chooses what the chips say and how they are coloured.
Filter	Shows only some lines, one direction or part of the day.
Stack: *n* unassigned	Vehicles only: how many trips have no bus yet.
OPTIMIZE **+ gear**	Lets RosterFlow make the blocks and duties. The gear chooses buses, duties or both.
Corners	Full screen: the KPIs folded away, more room for the rows.

The chart: rows on a time axis

Most of the screen is a chart with the hours of the day from left to right. In **Vehicles** each row is one bus's block; in **Drivers** each row is one duty.

- **On the left of each row** are a tick box (to tick the row for actions on several rows), sometimes a red **!** when the row breaks a rule, the line's colour stripe and the ID.
- **Hover over the ID** for the lock (Optimise leaves the row alone) and the pin (keeps it at the top).
- **Click the ID** to open the vehicle's card or the duty's panel.
- Under the last row is **+ New vehicle** (or **+ New duty**): drop trips there for an extra bus or a new duty.

The status bar

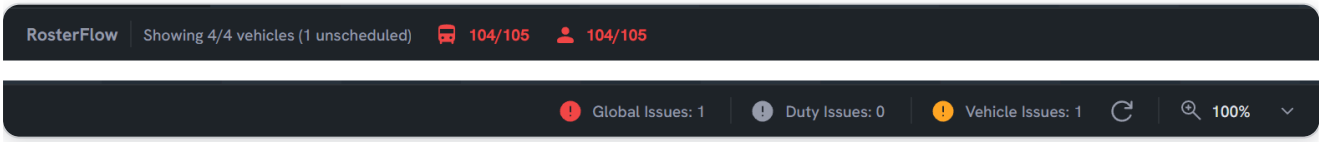


Figure 3.8 The status bar: what is shown, whether every trip is covered, the issues, validate again and the zoom.

- **Showing 4/4 vehicles (1 unscheduled)**: how many rows you see after the filter.
- **Bus icon 104/105**: trips with passengers assigned to a vehicle. **Person icon 104/105**: trips with passengers in a duty. The numbers are **red** until every trip has both.
- **Global Issues, Duty Issues, Vehicle Issues**: click one to open its list (chapter 11).

- **Round arrow**: validate again. **Magnifier with percentage**: the zoom (100 to 600%).

TIP: ZOOMING AND SCROLLING

Ctrl + mouse wheel zooms the hours in and out; Shift + mouse wheel moves along the day. That way you look at a busy rush hour up close without losing the overview.

02

PART II

Planning, step by step

The chain in practice. Each chapter is one tab, in the order you build a day: from a line's route to the driver who works the early duty on Monday.

4 Routes: the line itself

5 The Timeplan

6 Vehicles: building blocks

7 Drivers: building duties

8 The roster

CHAPTER 4

Routes: the line itself

Before a line can get a timetable it needs a route: the way the bus drives, the stops in the right order and the minutes between them. You do that on the Routes tab, without leaving RosterFlow.

What the Routes tab is

The **Routes** tab is Omsi-Hub's line editor in RosterFlow's look. A route you make or change here is saved in the same place as in the line editor itself, so the line editor, the Timeplan and the tenders all see the same route.



Figure 4.1 The layout of the Routes tab (sketch): the route bar at the top, the chosen direction's stops on the left, the map on the right, the running-time strip along the foot.

The four parts of the screen

1. The route bar

- **The route**: choose which line you edit, or start a **new route**: *click its stops on the map*.
- **The days** the route runs, as its kind of service has them, for example *Mon-Fri, Sat, Sun*, *Mon-Fri, school days*, *Sat, Sun* or *Mon-Fri, Sat, Sun, on demand*. The Timeplan makes its trips on these days.
- **The colour** of the route: the colour of its chips throughout the studio.
- **The two directions**, named after where they go: *To Waldfriedhof*, *To Grundorf Bahnhof*. A route without a way back is a **ring**: the bus goes round to its first stop.
- **Save**: a new route is confirmed and paid for; a change to a route in service is saved.

2. The direction's stops (left)

The stops of the chosen direction, in order. Drag a stop to another place in the order, or take it out. Below you see how many stops there are and how long the route is in kilometres (or miles). New stops go in after the stop you chose; click it again to add them at the end.

3. The map (right)

The route over the roads in its colour, with its stops and the **START** and **END** marks. Click a stop on the map to add it to the direction: the way to it is found over the roads. With **Search a stop** you type a name; then choose **Show it on the map** or **Add it to the direction**.

4. The running-time strip (foot)

Along the foot are the stops with their codes and the minutes from each stop to the next, plus the trip time. Those minutes come **from the roads** at first; click a number to change it (*typed in*). The Timeplan starts its running times from these minutes.

> **HOW IT WORKS: ONE SOURCE OF RUNNING TIMES**
> The minutes in the strip and the running times in the Timeplan are the same running times. A change here is then in the Timeplan, not saved yet: save it there. Press the roads'-minutes button twice, and the minutes from the roads replace the running times of *every* time band of that day group and direction.

Making a new line

1. **Start a new route**
   Choose a new route in the route bar. The stop list is empty.
2. **Click the stops**
   Click the first stop on the map (START), then the next, up to the last (END). The way between them is found over the roads by itself.
3. **Check the order**
   Drag stops in the list on the left into place if needed.
4. **Make the way back**
   Choose the other direction and click its stops, or leave it empty for a ring line.
5. **Choose days and colour**
   Set the days and the colour in the route bar.
6. **Confirm and pay**
   Save the route. A new line is confirmed and paid for: *Line 12 is the company's now*.
7. **Make the timetable**
   Go to the Timeplan tab. A line runs no tour until it has a timetable.

> **WATCH OUT**
> Save a route before you leave: the Timeplan always opens a line as it is *saved*.

> **ALSO VIA THE LINES PAGE**
> You can also draw a line of your own via the company's **Lines** page, in the line editor. There you confirm and pay for its route. With RosterFlow on, you then make its timetable in the Timeplan: the button *Open in the Timeplan* takes you straight there. A line of the map's own timetable you open once in the line editor to be able to plan it here.

CHAPTER 5

The Timeplan

Here a line gets its timetable: the minutes a bus needs from stop to stop, per part of the day, and the departure time of every trip. Every other step is built from these trips.

The bar at the top

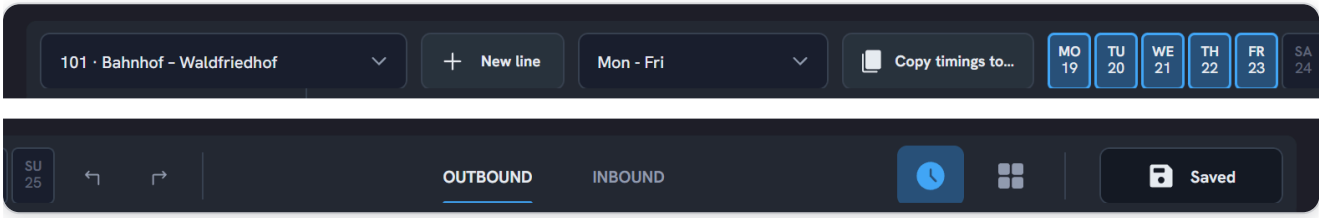


Figure 5.1 The Timeplan's bar.

PART	WHAT IT DOES
Line picker	The line you plan, such as *101 · Bahnhof – Waldfriedhof*.
+ New line	Draw a new line's route in the line editor, then make its timetable here.
Day group	*Mon - Fri*, *Saturday* or *Sunday*. Each day group has its own running times and trips.
Copy timings to...	Copies this day group's running times to other day groups.
Day strip MO 19 ... SU 25	The days of the coming week that run this timetable (outlined in blue).
Arrows	Undo and redo, within the Timeplan.
OUTBOUND / INBOUND	The direction you see.
Clock and grid	Switch between **Running times** (clock) and **Timetable** (grid).
Save	Writes the line's timetable. When it reads *Saved*, there is nothing to keep.

Running times: minutes per time band

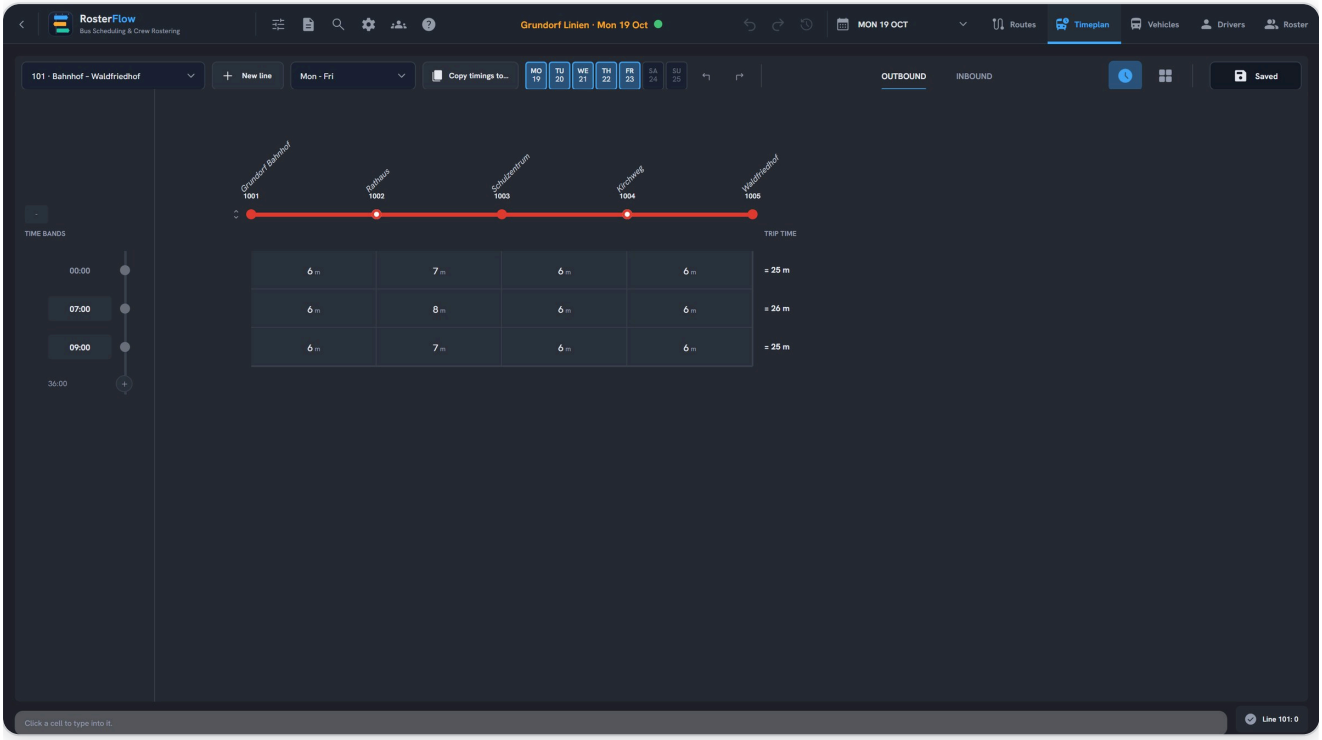


Figure 5.2 The running times of line 101, outbound, Monday to Friday: three time bands (from 00:00, 07:00 and 09:00) with the minutes between each two stops and the trip time on the right.

At the top you see the route as a coloured line with its stops, their names at a slant and their codes underneath. Below it a table: each **row** is a **time band**, each **column** the stretch between two stops, and each cell the minutes. On the far right is the **trip time**: the sum of the row.

Changing a running time

1. **Click a cell**
   The box becomes an input.
2. **Type the minutes**
   Such as `7`. `↑` and `↓` change it by a minute.
3. **Confirm**
   `Enter` keeps it, `Tab` moves to the next cell, `Esc` leaves without keeping.

Every trip leaving in that band moves along at once. That makes this the quickest way to make a whole day more realistic: make the rush hour a few minutes slower and all the rush-hour trips follow.

Adding, moving and removing time bands

- On the left is the **time bands** column with each band's start. Click a start time and type a new one.
- The **+** at the bottom adds a band; hover between two bands to put one in between (*Add a time band here*).
- Hover over a band for the button that takes it out.
- The day runs until **36:00**: times after midnight count on (01:30 at night is 25:30), so a night trip stays with its day.

Timepoints

The dots on the line at the top are stops. A **filled** dot is a **timepoint**, a **white** dot a plain stop. Click a dot to switch. In the timetable you can then show the timepoints only.

Copying running times to other day groups

Are Saturday's and Sunday's running times the same as the weekdays'? Choose **Copy timings to…**, tick the day groups and choose **Both directions** or **Only the direction shown**. Their trips follow the new running times. Save keeps them, Undo takes them back.

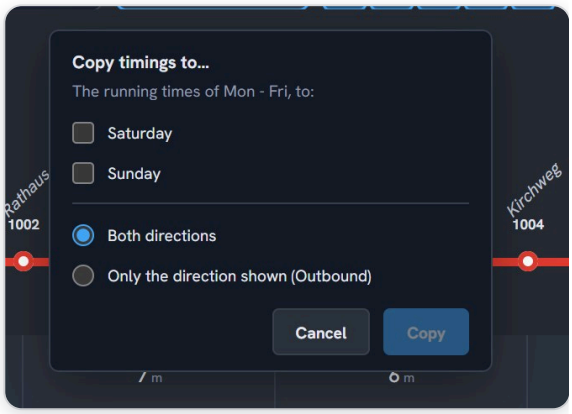


Figure 5.3 Copying running times to Saturday and Sunday.

The timetable: every trip a row

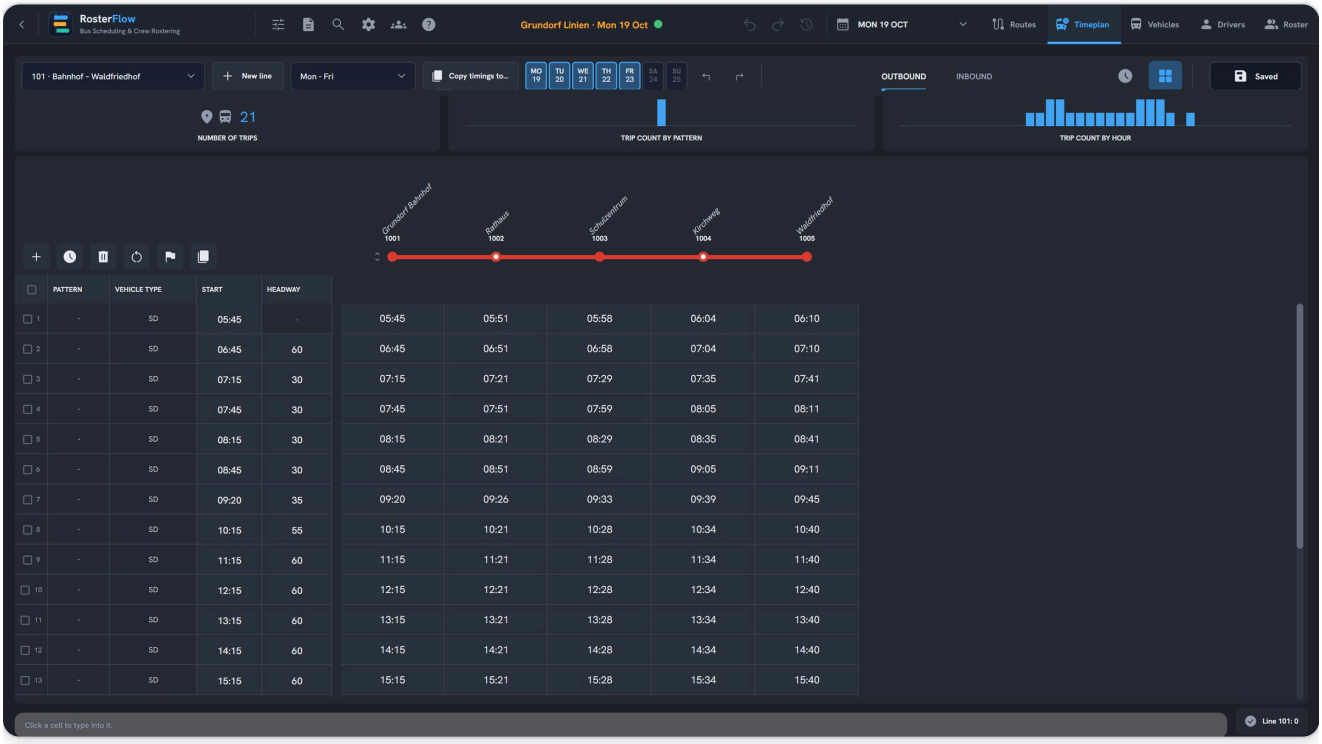


Figure 5.4 Line 101's timetable: the figures at the top, the tools below them and a row per trip with its pattern, vehicle type, start, headway and its time at every stop.

Click the grid icon on the right of the bar. Each row is now a **trip**: on the left its number, its **pattern** (- is the main pattern), its **vehicle type**, the **start** and the **headway** since the trip before; on the right its time at every stop.

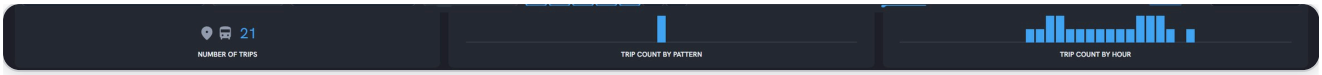


Figure 5.5 The three figures above the timetable: number of trips, trip count by pattern and trip count by hour. Click a figure to open its chart large; the arrows page to the next.

The tools

BUTTON	WHAT IT DOES
+	**Add a single trip**, after the one chosen, at its headway.
Clock	**Add trips by headway** from a control point (see below).
Bin	**Take the chosen trips out** (`Delete`).
Circular arrow	The chosen trips' times **from the running times again**: their typed times go.
Flag	Show **timepoints only** in the table.
Copy	**Copy to…** other days: this day, the trips chosen or a template.

Adding trips by headway



Figure 5.6 The form for trips by headway. On the right you already see how many trips it makes; the tick makes them.

1. **Click the clock tool**
   A form appears above the table.
2. **Choose the control point**
   The stop the times are for, usually the first stop.
3. **Fill in start and end time**
   The first and the last departure, such as `06:00` and `22:00`.
4. **Fill in the headway**
   The minutes between the buses, such as `15`. Several values, such as `10,20`, are taken in turn: a bus then leaves alternately after 10 and after 20 minutes.
5. **Click the tick**
   66 trips added. The cross closes the form without making anything.

> ✓ **TIP: PEAK AND OFF-PEAK**
> Build a day in parts: first 06:00–09:00 every 10 minutes, then 09:00–15:00 every 20, then 15:00–18:00 every 10 again and finally the evening every 30. Four steps and you have a realistic timetable with a peak.

Choosing and editing trips

- **Click a trip's number** to choose it. `Shift`+click chooses a range, `Ctrl`+click adds one or takes it off, `Ctrl`+`A` chooses every trip, `Esc` none.
- With several trips chosen a **selection bar** appears: move them a number of minutes **earlier** or **later**, or give them another **pattern**.
- **Click a time** in the table to type it by hand (such as `07:45`, up to 36:00). A time typed by hand turns **orange**; the circular arrow puts it back to the running times.

Copying to other days, and templates

With **Copy to…** you put trips onto other day groups. First choose **what** to copy: the whole day, the trips chosen or a saved **template**. Then tick the day groups and choose how:

- **Replace their trips**: the days ticked run these trips, with their running times, instead of their own.
- **Add to their trips**: the trips go beside the days' own. One there already (same way, same minute) is not copied twice.

At the bottom you keep the day or the trips chosen **as a template** with a name, such as *School days*, to copy onto any day later. Templates can be renamed and thrown away (click twice to confirm).

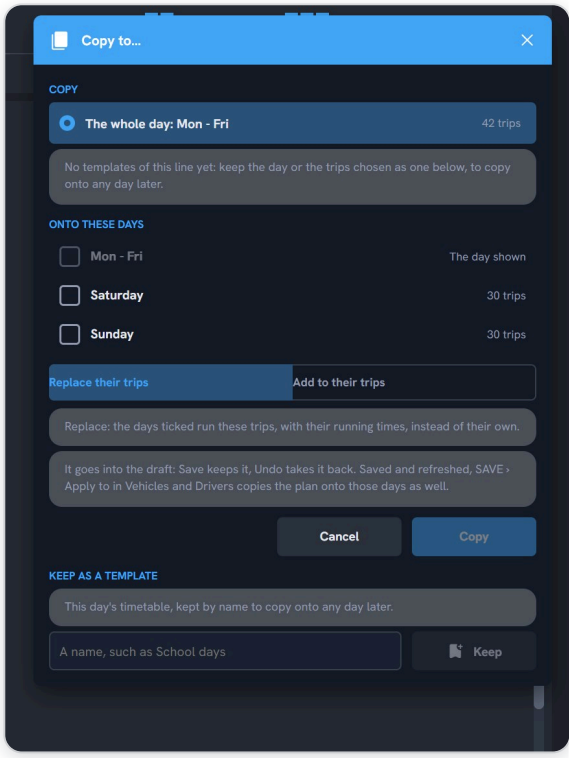


Figure 5.7 Copy to… with the choice of what, where to and how, and keeping a template.

Checking: the timetable must be valid

At the bottom right is a button with the number of issues of this day group, such as *Line 101: 0*. Click it for the list. A valid timetable has none. The possible issues:

ISSUE	FIX
Trips leave at the same minute as another trip of their pattern	Take one of the two out, or move it.
Trips arrive after the trip that leaves after them	The running time is too long for the headway; check for typing errors in the times.
Running times below zero / of zero minutes	Type a positive number of minutes in the running times.
Patterns without stops	Give the pattern its stops in the line editor or in Routes.
Trips run past 36:00	Let the last trips leave earlier.

Save, then Refresh

This is the most important action in the whole Timeplan, and the most often forgotten:

1. **Save the timetable**
   Click **Save** at the top right. If RosterFlow asks you to mend something first, the timetable isn't valid yet: fix the issue (for example two trips in the same minute) and try again.
2. **Read the status bar**
   It says **UNSYNCED – Refresh**: the timetable changed since this line's blocks were made.
3. **Click Refresh**
   The buses' blocks and the duties of this line are made anew for the new trips.
4. **Go to Vehicles and Drivers**
   Check the result, refine it and save there.

> **WATCH OUT**
> Refresh throws away the saved blocks and duties of that line on that day group and makes them anew for the new trips. Changes made by hand to that line in Vehicles and Drivers are lost. So finish the timetable first, and only then start fine-tuning the blocks and duties.

> **LEAVING WITHOUT SAVING**
> If you try to leave the Timeplan with unsaved changes, RosterFlow warns you: *Changes not saved: press again to leave without them.*

CHAPTER 6

Vehicles: building blocks

In Vehicles you share the day's trips out among your buses. Each row is one bus's block. This is where you decide how many buses you need and how many empty kilometres they drive.

Reading a block



Figure 6.1 Four blocks. Block 101/02 also runs two trips of line 205 (interlining); 102/01 has a red "!": it breaks a rule.

A block starts on the left with the **pull-out** (T-mark and grey-blue box), then come the **trips** as coloured chips, and it ends with the **pull-in**. Arrows in a row mark where drivers change. A small dark box is a **deadhead**. The ID, such as **101/02**, is the line and the block's number.

> **WHAT MAKES A GOOD BLOCK?**
> A good block keeps the bus busy: short layovers, few and short deadheads, and as few buses out at once as possible. Watch **Vehicle efficiency** (higher is better) and the number of **blocks** (lower is better).

A trip's card and a vehicle's card

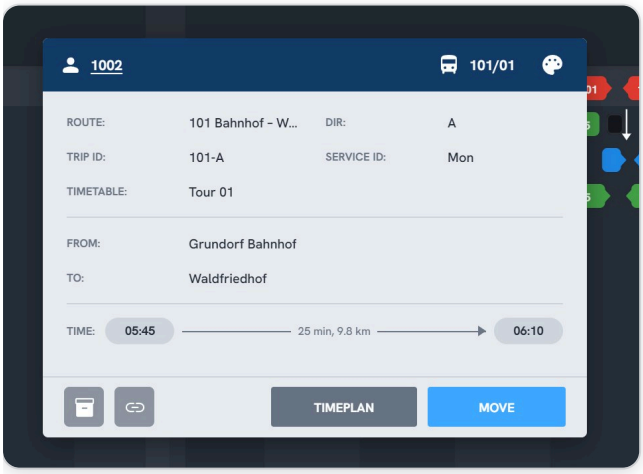


Figure 6.2 A trip's card: route, direction, trip ID, service ID, block, from and to, times, duration and kilometres.

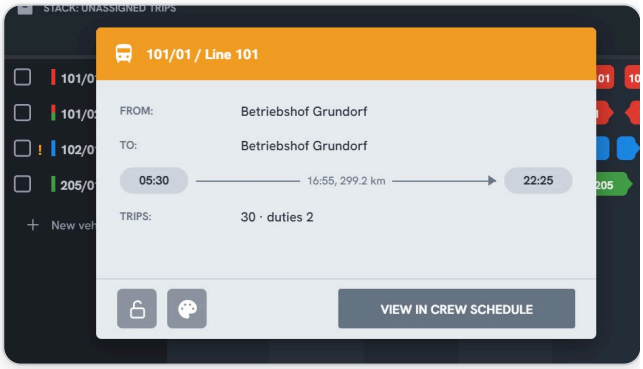


Figure 6.3 A vehicle's card: start and end, the length of its day in hours and kilometres, the number of trips and duties.

Click a chip to choose the trip and open its card. At the top you see which duty and block it is in. At the bottom: **Unassign** (to the stack), **Link**, **Timeplan** (the trip in the Timeplan) and **Move** (pick it up). **Click an ID** for the vehicle's card, with buttons to lock it, choose the line colour and **View in crew schedule** (to Drivers).

Moving a trip to another bus

This is RosterFlow's basic action. You pick up a trip, RosterFlow shows where it can go and what that costs, and you choose.

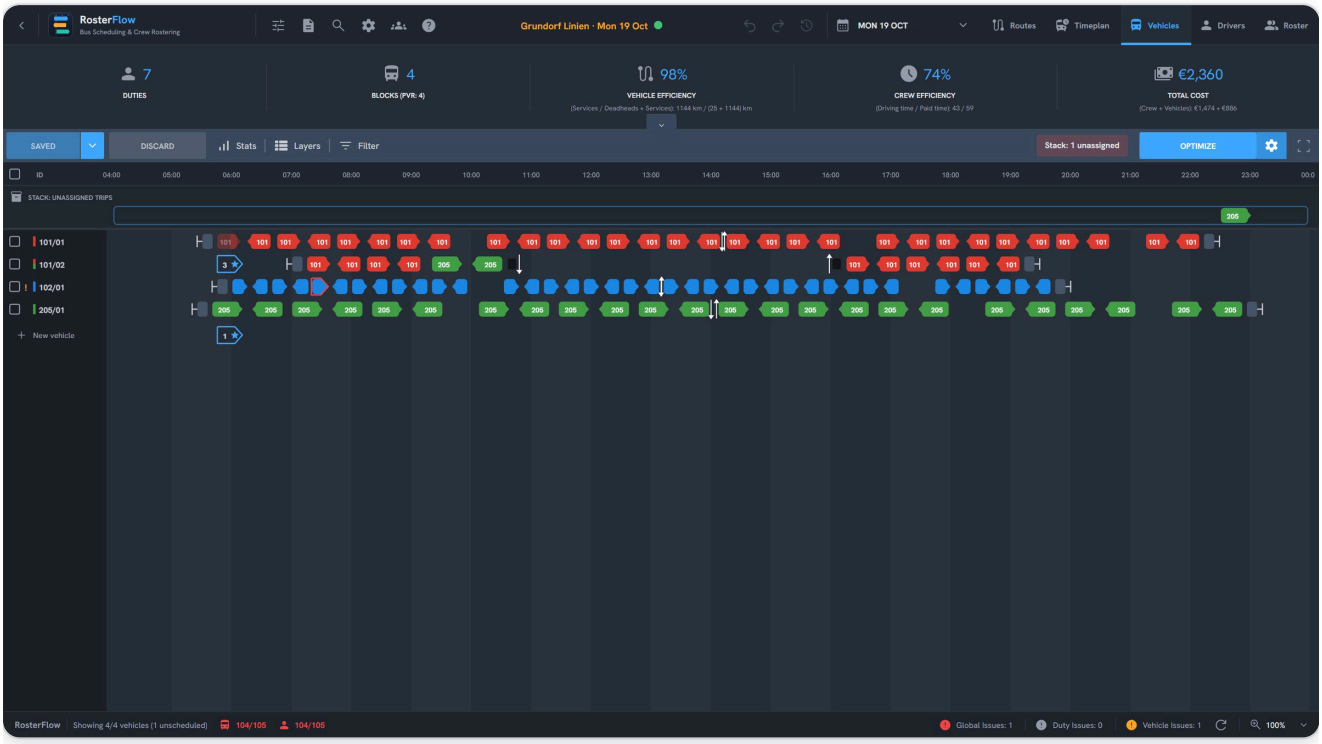


Figure 6.4 A trip picked up (left, on 102/01): every bus it could go to shows a drop slot with stars, and so does New vehicle.

1. **Pick up the trip**
   Drag a coloured chip, or double-click it, or choose **Move** on its card. A **drop slot** appears on every bus it could go to.

2. **Read the stars and the cost**
   Hover over a slot. Five stars is the cheapest choice. **Blue** means no rule is broken, **red** that a rule does break. The card shows the **cost difference**.

3. **Drop it**
   Click a slot, or let go of the chip over it. Esc puts it back.

4. **See the figures change**
   The green or red differences beside the KPIs show what your move did.

5. **Not good? Undo it**
   Ctrl + Z puts the trip back on its bus.

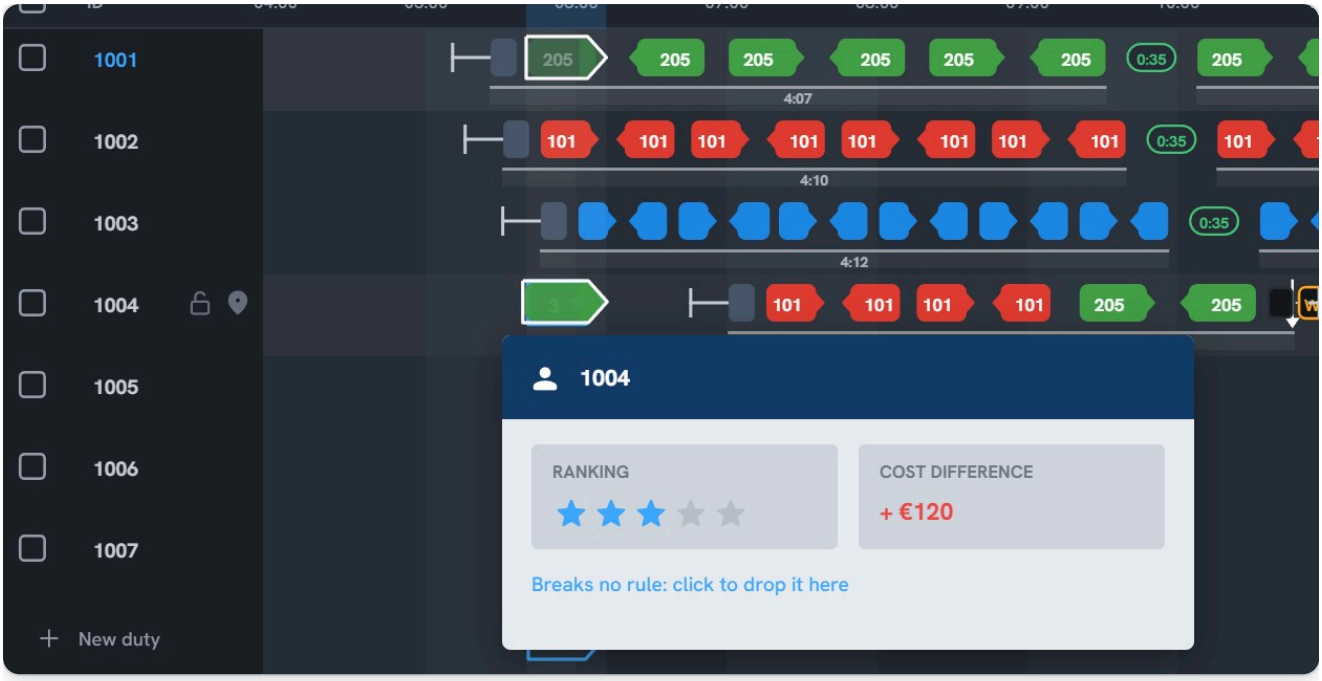


Figure 6.5 The card over a drop slot: three stars, cost difference + €120, and “Breaks no rule: click to drop it here”.

> **HOW IT WORKS: WHY NO SLOT?**
> A bus only gets a slot if the trip fits it in time. If the trip overlaps another trip of that bus, or the bus can't be at the first stop in time, there is no slot. Red is allowed: it may be done, but the day gets an issue.

Several trips at once

- Shift +click chooses more trips of one row, as a range.
- Ctrl +click adds one trip to the choice, or takes it out.
- The ← → arrow keys go to the trip before or after; with Shift the choice grows. ↑ ↓ jump to the row above or below.
- Pick up one of the trips chosen, and they all go along.

A trip's menu

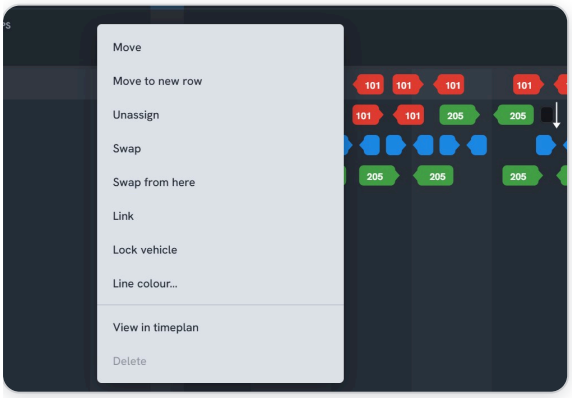


Figure 6.6 Right-click a chip.

MENU ITEM	WHAT IT DOES
Move	Pick up the trip, like a drag.
Move to new row	The trips chosen onto a new bus.
Unassign	The trip to the stack: it waits there for a bus.
Swap	Swap the trips chosen with what another bus runs over their time. Click that bus next (Esc : no swap).
Swap from here	Swap everything from this trip on with another bus's day from then on. Click that bus next.

MENU ITEM	WHAT IT DOES
Link / Unlink	Trips chosen stay together on one bus; Optimise keeps them so.
Lock vehicle	Optimise leaves this vehicle alone.
Line colour…	Choose the line's colour.
View in timeplan	To this trip in the Timeplan.
Delete	Greyed: trips are deleted in the timetable.

✓ **SWAPPING IS POWERFUL**

Swap from here is ideal when two buses cross each other halfway through the day: one takes over the other's day from that point on. That way you fix a block that is too long or too short in one click.

The stack: trips without a bus

Above the rows is the **stack** (*Stack: unassigned trips*). Trips no bus drives wait there: new trips, or trips you unassigned. As long as something lies on the stack, the bus number in the status bar is red and RosterFlow reports the global issue *n trips are not assigned to a vehicle*.

- To **put a trip on the stack**: choose **Unassign**, press `Delete`, or drag it to the stack (*Drop here to unassign*).
- To **give a trip from the stack to a bus**: click it and then one of the slots that appear.
- Or let OPTIMIZE empty the stack.

An extra bus, and pinning rows

Drop trips on **+ New vehicle** under the last row to give them to an extra bus. A row you need often you **pin to the top**: hover over the ID and click the pin, or choose the row and press `P`. With `G` you jump to a bus or duty by its ID.

Working with ticked rows

Tick rows with the box before the ID (or all of them with the box in the header). A selection bar appears with:

- **Lock** and **unlock** the rows ticked;
- **Unassign the ticked vehicles' trips to the stack**, to plan them anew;
- **Copy to…** other days (also with `Ctrl`+`C` and then `Ctrl`+`V` on another day).

Copying rows to other days

Made a perfect block on Monday? Tick it, choose **Copy to…** and tick the days. For each day you see whether it is the **same day type**. Choose **Every week** or **This date only**, and click **Copy**.

Only the trips that match there go along: **same line, way, departure and stops**. The rest are listed. It goes into your draft: SAVE keeps it, Undo takes it back.

> **WATCH OUT**
> A locked vehicle or duty on the target day keeps its trips. Unlock it there and copy again to take them.

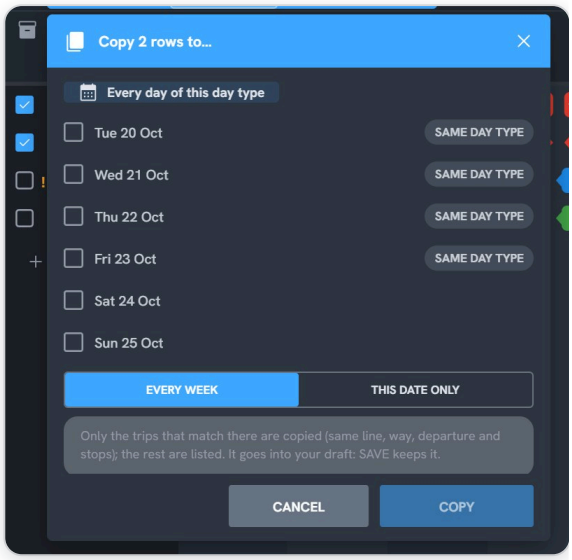


Figure 6.7 Copying two rows to Tuesday to Friday.

Cutting duties, already in Vehicles

Duties are cut from the buses' days, and that can be done here already. Hover over the gap between two trips. Where a **blue line** appears, a driver can take over: click to cut there (*Click to cut the duty here: a driver takes over at Waldfriedhof*). The arrows that appear mark your cut; click the arrows to join the two duties again. Chapter 7 goes further into this.

Giving a line its own colour

Right-click a trip and choose **Line colour…** (or the palette on the card). Click a colour in the row, or any colour with the picker or a hex code. Every chip of the line takes it at once. **Automatic** gives the line its old colour back.

> **TIP**
> Give lines that often run on the same bus clearly different colours. Then you see interlining at a glance.

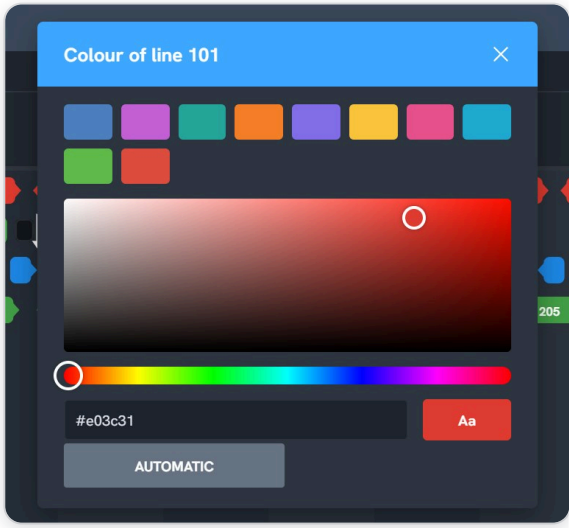


Figure 6.8 The colour of line 101.

CHAPTER 7

Drivers: building duties

In Drivers each row is one driver's working day. This is where you decide how many drivers the day needs, when they take their breaks and whether their duties keep the rules.

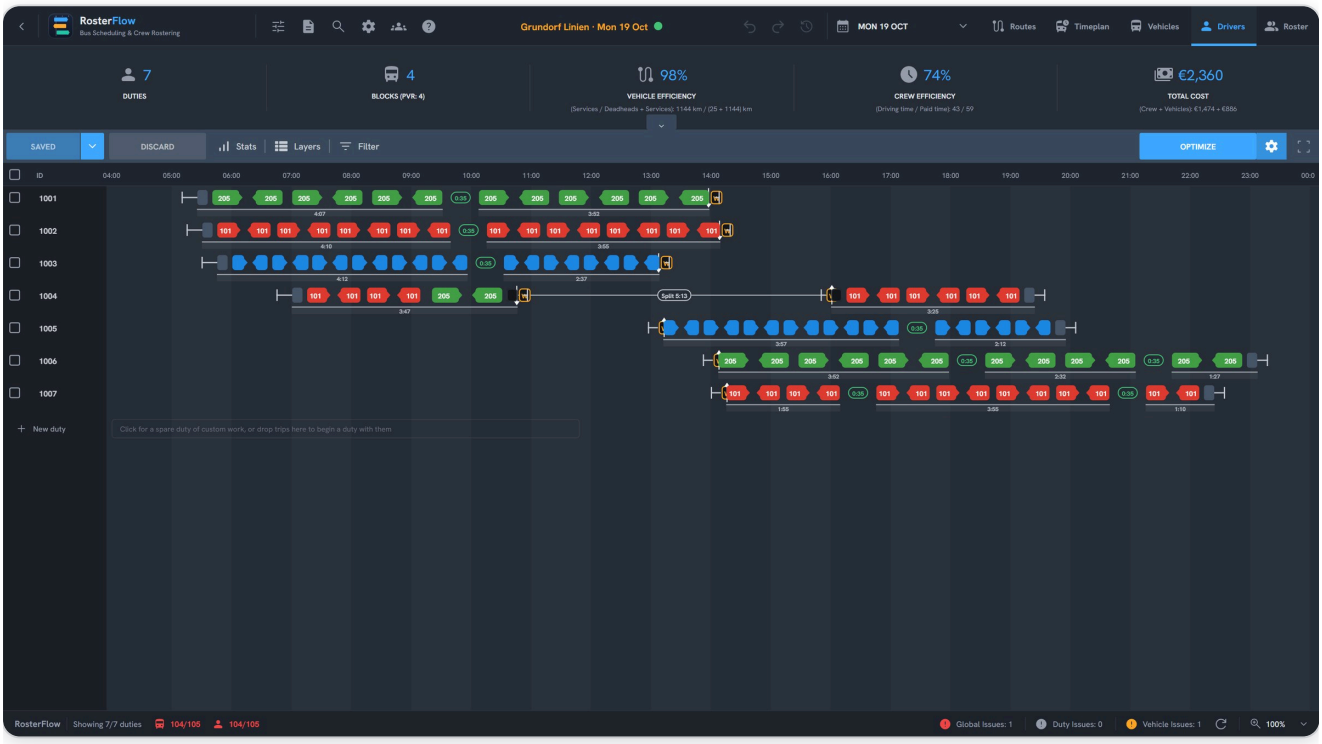


Figure 7.1 Seven duties. Duty 1004 is a split duty (Split 5:13). Under each duty its pieces of work with their length; in the gaps the breaks (0:35).

Reading a duty

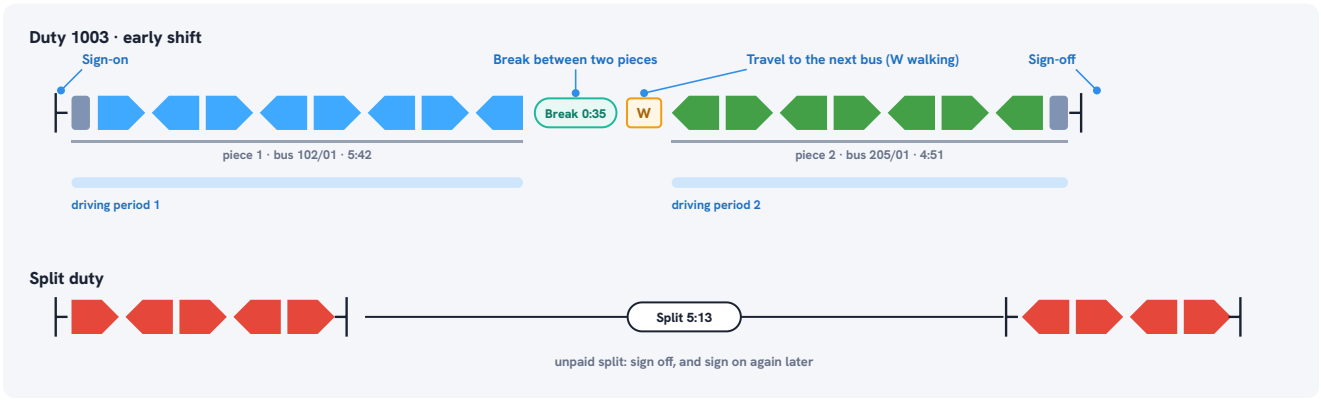


Figure 7.2 A duty from left to right: sign-on, first piece of work, break, travel to another bus, second piece, sign-off. Under the duty the band of driving periods.

- A duty begins with the **sign-on** and ends with the **sign-off**: the T-shaped marks.

- In between are the **pieces of work**: the parts on one bus. Under the row each piece has a line with its length, such as *4:07*.
- Each gap says what it is: a **break** (green pill with its length), a **meal**, or a **split** of a split duty (*Split 5:13*).
- An amber box with **W**, **P** or **R** is the driver on the way to the next bus: walking, public transport or staff car.
- Under the duty lies a band of **driving periods**: the time between breaks long enough to end a driving period. Hover over it for the details, such as *within the Continuous Break limit*.

> **BREAK OR STANDING TIME?**
> Not every gap is a break. A gap shorter than the threshold in **Preferences › Duty Breaks › Show a gap as a break from** (0:15 by default) is *standing time*: no break, it ends no driving period and counts for no rule.

Cutting and joining

Duties come from cutting blocks. A cut can only be made where a driver can get on or off: at a **relief point** (the depot, a terminus, or a stop on the way you added in Preferences).

1. **Go to Vehicles**
   Duties are cut from the buses' days, so you start here.
2. **Cut at a terminus**
   Hover over the gap between two trips. Where a blue line appears, click to cut there.
3. **Look at it in Drivers**
   There is one duty more now: the second driver's day starts where you cut.
4. **Join it again**
   In Vehicles, click the arrows where the drivers change. The two duties become one.

Relief halfway along a trip

If you added a stop on the way in **Preferences › Relief Points**, drivers can also change in the middle of a trip: *Click to cut the duty here, on the way: a driver takes over at Rathaus*. Such a relief is called **Relief on the way**. A trip can have only one, and the stop must lie between the trip's first and last stop. *Join the two duties* lets one driver work the trip whole again.

Moving trips between duties

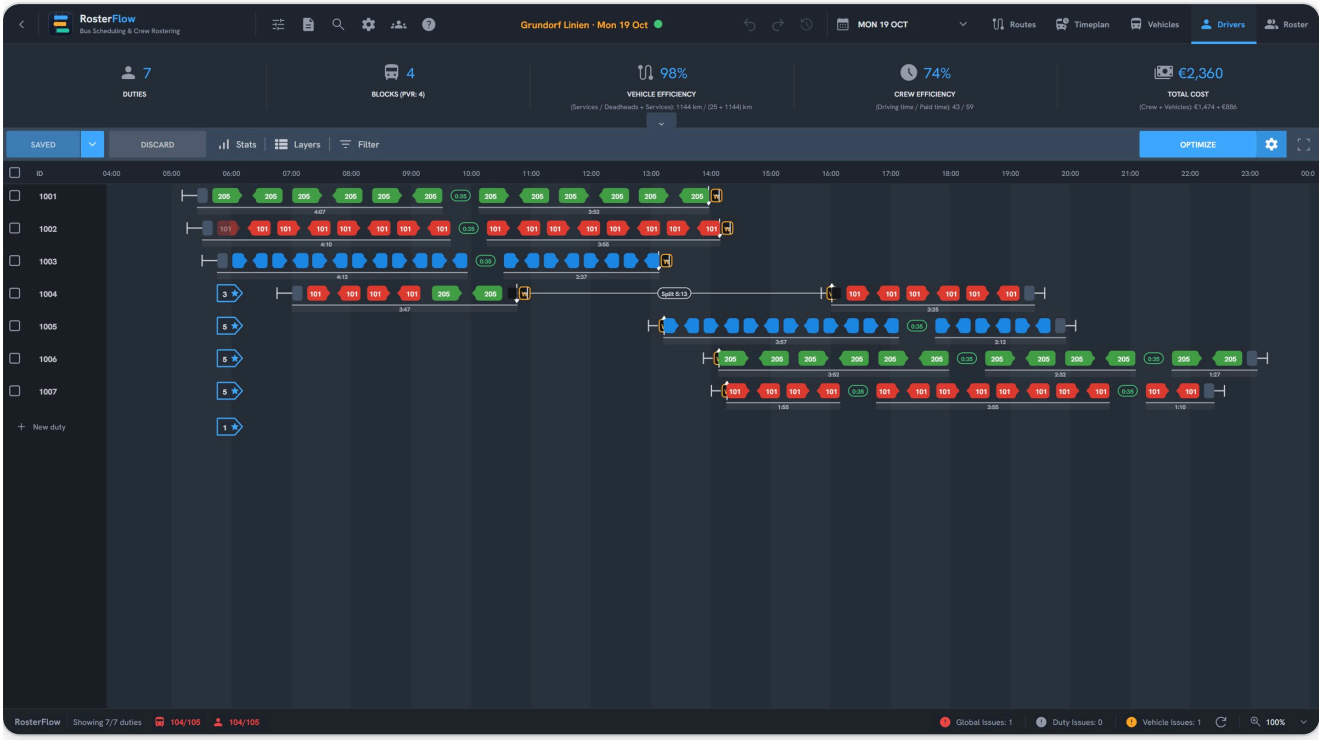


Figure 7.3 A trip picked up in Drivers: every duty that could take it, and New duty, shows a slot.

In Drivers you move work between drivers, while **the bus stays the same**:

- **Drag a chip** (or double-click) onto another duty: that duty's driver drives the trip now, on the same bus, and the first duty is cut around it.
- `Alt` **+ drag** takes the **whole piece of work** on that bus along.
- `Shift` **or** `Ctrl` **+ click** chooses more trips, also of another duty.
- When several duties are **ticked**, a chip of one of them takes all of their trips along: each duty moves the same number of rows up or down.

> **NO SLOT?**
> Then the duty would break a rule (for example get too long), or the drivers can't change at a terminus there. Take a shorter piece, or look in Preferences for the rule that stops it. When it can't be done, RosterFlow says so, for example: *The driver cannot be there in time: the work before it ends at another stop, too shortly before.*

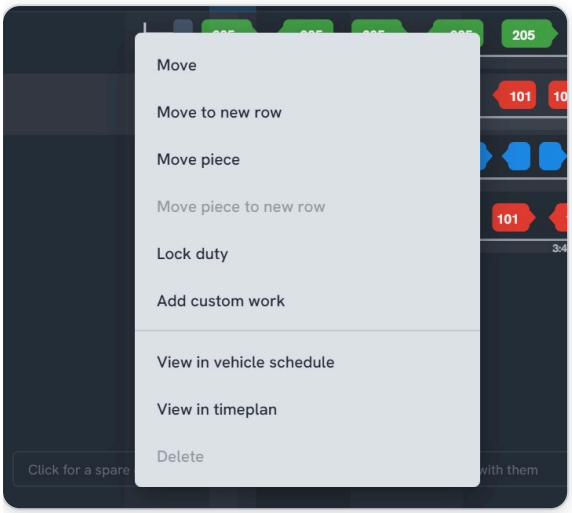


Figure 7.4 A trip's menu in Drivers.

The menu in Drivers

Move	Pick it up.
Move to new row	The trips chosen a duty of their own.
Move piece	Take up the whole piece of work (like Alt +drag).
Move piece to new row	Make the piece a duty of its own.
Lock duty	Optimise leaves it.
Add custom work	Spare, office, union or training time in the duty.
View in vehicle schedule	To Vehicles.
View in timeplan	To the Timeplan.

A duty's panel

Click a duty's ID on the left to open its panel. At the top are its number and its **type** (such as *Early shift*), with the lock and a cross.

- **Duty type**: *Automatic* (the first type in Preferences › Duty Types the duty fits) or a type you choose.
- **The figures**: sign-on and sign-off, spread, work time, driving time, platform time, splits and changeovers.
- **Paid time**: set it by hand with – and +; otherwise it is automatic.
- **Sign-on / Sign-off**: this duty's minutes for signing on and off.
- **Sign on at / Sign off at**: as the company has it (*Where the bus is*) or a relief point of its own.
- **Pieces of work**, **Breaks** and **Driving periods**: what the duty is made of.
- **Custom work**: choose the kind and the times and click **Add**.
- The duty's **Issues**, and for a spare duty **Delete duty**.

> **TIP**
> Lock a duty that is just right with the lock at the top of the panel. Optimise then leaves it as it is.

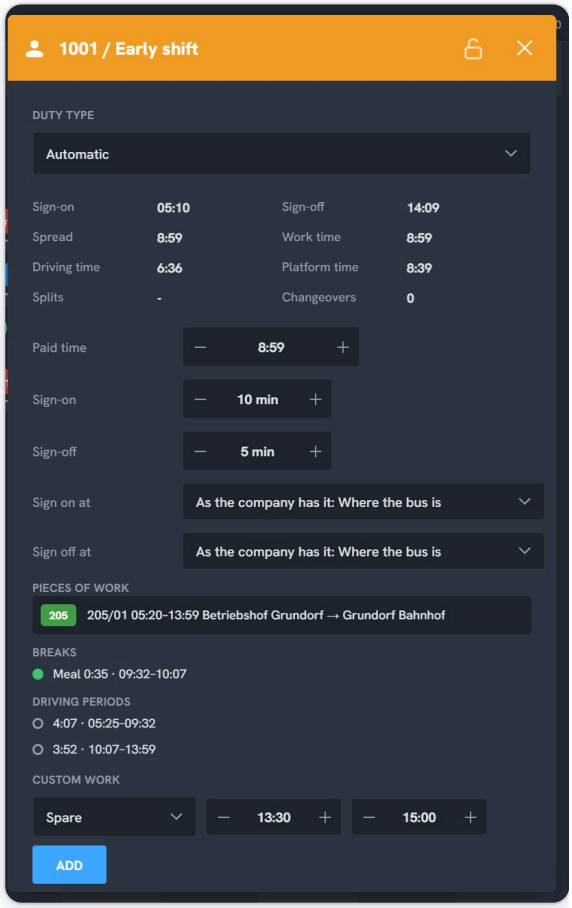


Figure 7.5 Duty 1001's panel.

Building a duty by hand

1. **Open the Drivers tab**
   Each row there is one driver's working day.
2. **Pick up a piece of work**
   Drag a chip of a short duty with Alt, or choose **Move piece**: the whole part on that bus goes along.
3. **Drop it on another duty**
   Click a slot: the two become one duty, with a break in between. No slot? Then the duty gets too long: take a shorter piece.
4. **Or start a new duty**
   Let go of trips or a piece on **+ New duty** at the bottom to begin a duty with them.
5. **Check the rules**
   Open the duty and look at Issues. A red "!" before the row means something is wrong.

Spare duties and spare lines

Not every driver has to drive trips. Click **+ New duty** for a **spare duty** of custom work. A spare duty takes no trips, but stands by.

- With **Add spare line** you put another spare line on the day. Drag an edge to change its start or end, or drag it whole to move it. A click opens its card: the times can be typed there, and you give it a name.
- **Stands by at**: where the spare driver signs on and stands by. Service control sends them from there.
- During the day, the dispatcher and service control use spare lines within their hours, for example: *The driver of spare line R1 covers them.*

✓ **WHY SPARES?**
A bus breaks down, a driver is late: without a spare, trips are dropped and you pay a penalty. One or two spare lines in the peak make your day much more robust.

Duties back to the stack

Tick duties and choose **Every trip of the duties ticked back to the stack** in the selection bar, to plan them anew. RosterFlow asks first; Undo (Ctrl + Z) puts them back. Trips whose vehicle or duty is locked stay where they are.

CHAPTER 8

The roster

The Roster tab links your planning to real people and real buses: which driver works which duty, and which bus runs which tour, for every day of the coming week.

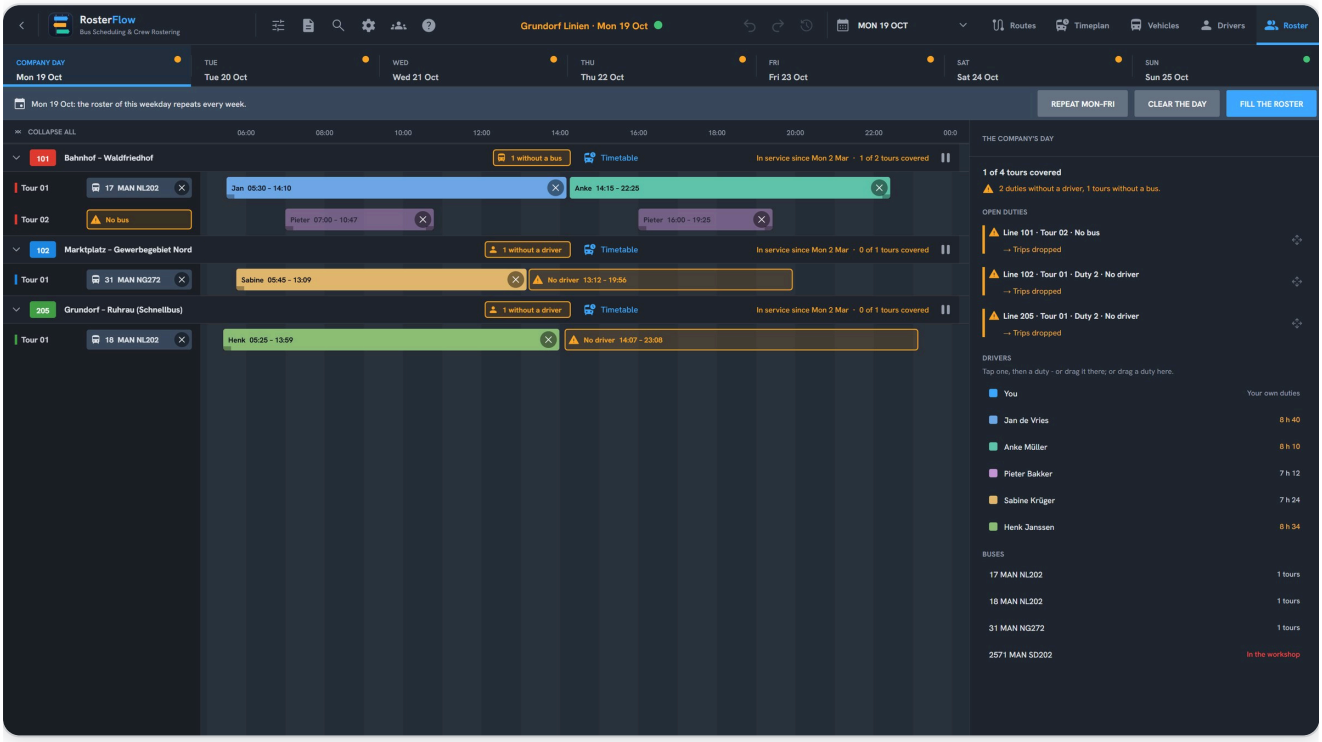


Figure 8.1 Monday's roster: per line the tours (Tour 01, Tour 02) with their bus and their duties, in the driver's colour. On the right the day, the open duties, the drivers and the buses.

Reading the roster

- **The week at the top**: the company day and the six days after it. The dot on each day shows whether it is covered. Your plan for a weekday repeats every week.
- **A heading per line**: the line's number and name, warnings such as *1 without a bus* or *1 without a driver*, a link to the timetable and how many tours are covered. Click the arrow to fold a line away.
- **A row per tour** (a tour is a block as the roster calls it): on the left its bus (with a cross to take it off), then the duties on the hours in the driver's colour. An orange box *No bus* or *No driver* is still open.
- **On the right**: how many tours are covered, the **open duties** with what happens if they stay open (*Trips dropped*), your **drivers** with their hours and your **buses** with their number of tours. A bus in the workshop is listed too.

Assigning drivers and buses

1. **Choose a duty or tour**
   Click a duty or tour in the chart to see who could take it.
2. **Choose the driver or bus**
   Click a driver or bus in the list on the right. Or drag a driver or bus from the list onto a duty or tour.
3. **The other way round works too**
   Tap a driver first and then a duty, or drag a duty onto a driver.

You are at the top of the drivers: plan yourself on a duty to drive it yourself. On the company day you can start a duty straight away in the company's duty launcher.

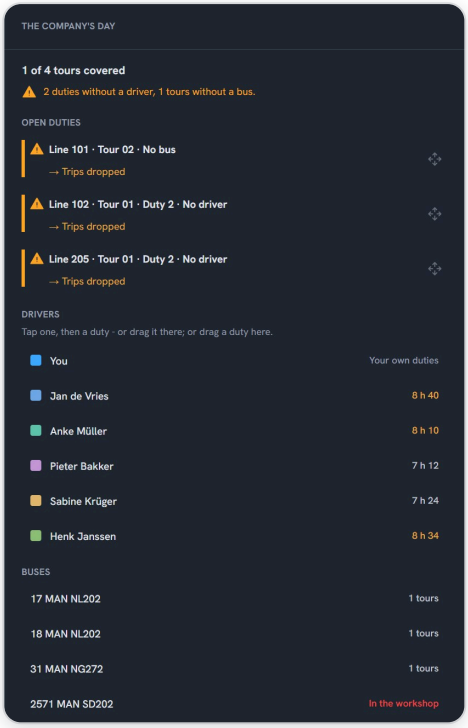


Figure 8.2 The side list: the day, open duties, drivers and buses.

Fill, clear, repeat

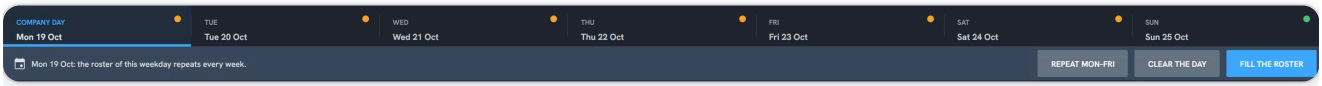


Figure 8.3 The week and the roster's three tools.

Fill the roster	Gives free drivers and buses to everything still open.
Clear the day	Takes every driver and bus off the day.
Repeat Mon–Fri	Copies this day's roster to the other days.

Taking someone or a bus off

Click the cross by a driver or bus, choose it and press **Delete**, or drag it back to the list (*Let go here to take the driver off*). With several chosen, you take them off at once. A right-click gives more:

MENU ITEM	WHAT IT DOES
Remove driver / Remove bus	From this duty or tour, for this day.
Give to… / Change bus…	Choose another driver or bus.
Show in Drivers	To this duty on the Drivers tab.
Remove from all duties today	Take the driver off all their duties of this day (likewise for tours and buses).

MENU ITEM	WHAT IT DOES
Give to an agency driver	An agency driver works the whole duty, for an amount per week shown beside it.
Cancel the rental bus	Cancel a rented bus: its amount for the day is not paid.

WHEN IT CAN NO LONGER BE DONE

On the company day itself the facts count: a duty that has begun can no longer lose its driver; a tour under way keeps its bus until it is back in the depot; what was driven stays. A duty or tour left without a driver or bus today has its trips **dropped**, with a penalty.

ROSTER AND DRAFT

The roster works with the *saved* duties. If RosterFlow has unsaved changes for this day, the tab says so: save them in Vehicles or Drivers first to plan with them here.

03

PART III

Refine

A planning that works is only the beginning. Here you make it cheaper, fairer to your drivers and valid for the whole week, and learn to set your company's rules yourself.

9 Optimise

10 Save, discard and other days

11 Finding and fixing issues

12 Preferences: your company's rules

13 Keeping the overview

14 Printing

15 RosterFlow in the game and planning together

16 Learning in the studio

CHAPTER 9

Optimise

With one button RosterFlow makes the blocks and duties itself, as cheaply as possible within your rules. The result is a proposal: you decide whether to take it.

What Optimise does

OPTIMIZE is on the toolbar of Vehicles and Drivers. It builds the day anew by the **Preferences**:

- **The blocks**: as few buses and empty kilometres as possible, with interlining where you allow it.
- **The duties**: the blocks cut at relief points and paired again into working days that keep every **hard** rule, with the cost plus the penalties for **soft** rules as low as possible.

Locked vehicles and duties stay as they are, and **linked** trips stay together on one bus.

Step by step

1. **Lock what must stay**
   A duty you made perfect by hand, or a block a particular bus must run: lock it (the lock by the ID).
2. **Choose what to optimise**
   The gear beside the button opens the settings: **Optimise vehicles (the blocks)** and **Optimise duties (cut and paired)**. Both are on by default.
3. **Click Optimize**
   RosterFlow works it out and shows the result as a **proposal**: *Optimised proposal: the KPIs show it against your planning.*
4. **Compare the figures**
   The KPIs show the proposal against your planning: green where it is better, red where it is worse.
5. **Accept it or keep yours**
   Accept puts the proposal in your draft; **Keep mine** forgets it.
6. **Save**
   The accepted proposal is in your draft: SAVE keeps it.

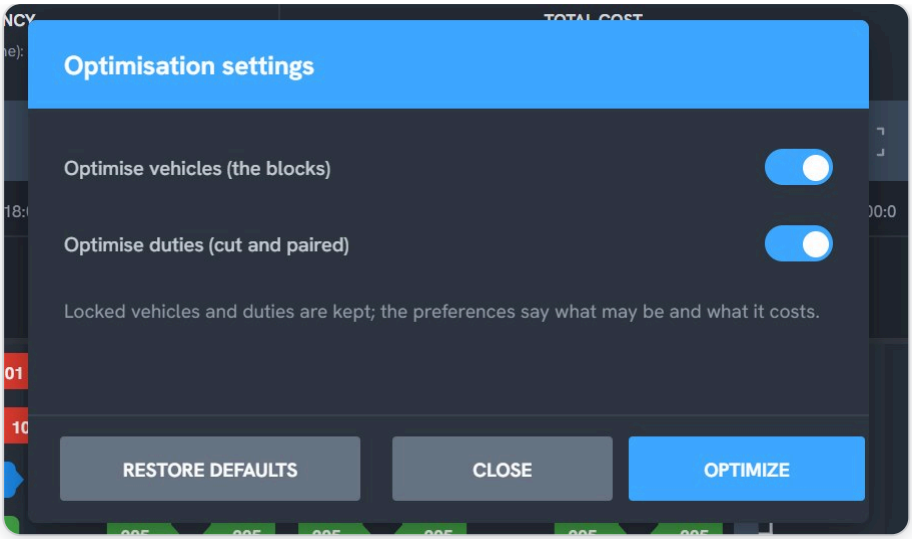


Figure 9.1 The Optimisation settings (gear). Restore defaults switches both on again.

What RosterFlow tells you

MESSAGE	WHAT IT MEANS, AND WHAT YOU DO
Optimised: 7 duties on 4 vehicles, €2,300 a day (your planning: €2,360)	A better plan was found. Compare and accept.
No better plan found: your planning already costs the least by the preferences.	Well done. If you want something else anyway, change the preferences (for example the costs or a rule).
Nothing to optimise: no trips run on this day.	Make a timetable for this day first.
Optimise makes neither the vehicles nor the duties now	Switch one on in the settings (the gear).
Hard rules broken: *n*, against *m* in your planning.	Even the proposal can't keep everything. Read the reason in the next message.
***n* duties can't keep every hard rule, because of: …**	The rules are too strict for the trips. Relax the rule named, or add relief points.
***n* vehicles have no relief point where drivers may change**	Allow relief at at least one terminus of those lines in Preferences › Relief Points.

Optimising smartly

STRATEGY 1

Rules first, then the button

Optimise is only as good as your Preferences. Set your duty types, breaks and costs first (chapter 12); then the button gives you something that suits your company.

STRATEGY 2

Buses only

Switch **Optimise duties** off to improve only the blocks. Handy when your duties are already good and you just want to save a bus.

STRATEGY 3

Lock and repeat

Optimise, lock the rows you like, adjust the rest by hand and optimise again. Each round, the good stays.

STRATEGY 4

After changing the rules

Save your Preferences and RosterFlow shows what the new rules break now, with the button **Optimise them**. One click and the day fits again.

CHAPTER 10

Save, discard and other days

Save makes your draft the company's planning. With the arrow beside it you put the same planning on other days at once, as a pattern for every week or as an exception for one date.

Save and discard

- SAVE makes the draft the company's planning: *Saved: the roster, the day and the game run this planning now.* The dot turns green.
- DISCARD forgets your changes: *Discarded: back to the saved planning – the locked vehicles and duties stay as they are.*
- A plain save holds for **this weekday, every week**: *Saved for this weekday: its planning holds every week.*

Save and apply to other days

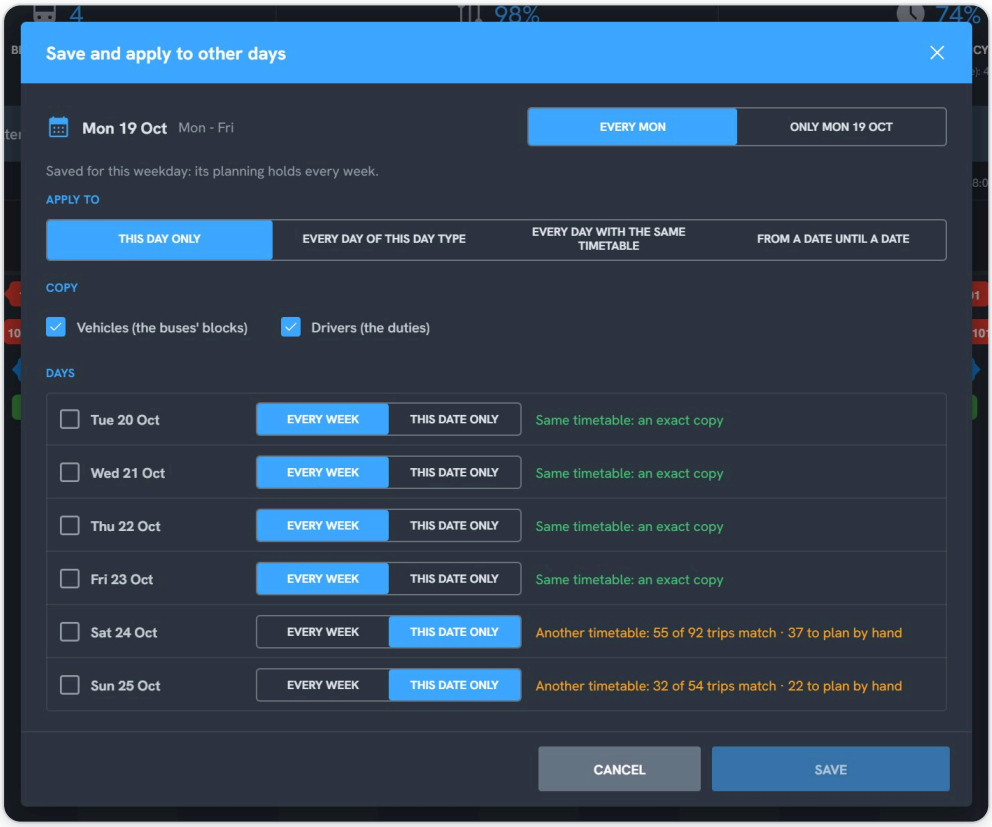


Figure 10.1 The "Save and apply to other days" window. Tuesday to Friday run the same timetable (exact copy); Saturday and Sunday another one (only the matching trips).

Click the arrow beside SAVE. The window has four parts:

1. The day shown: every week or this date only

At the top the day you plan, with its day type. Choose **Every Mon** (the planning holds every week) or **Only Mon 19 Oct**: *Saved for this date only: its weekday keeps the planning it had (an exception for one day).*

2. Apply to

This day only	Copy nothing, just save.
Every day of this day type	For example every weekday when you plan a Monday.
Every day with the same timetable	Every day that runs exactly the same trips.
From a date until a date	A range, *until* and including the second date: these dates only, afterwards each runs its weekday's planning again.

3. Copy

Tick what is copied: **Vehicles (the buses' blocks)**, **Drivers (the duties)** or both.

4. Days

Every day of the coming week, with a tick box, the choice **Every week** or **This date only**, and what will happen:

- **Same timetable: an exact copy**: everything fits.
- **Another timetable: 55 of 92 trips match · 37 to plan by hand**: only the trips that match (same line, way, departure and stops) are taken over. The rest run as their timetable has them and are listed, to plan by hand.
- *Its own changes are replaced by the copy*: that day already had changes of its own.

After copying

RosterFlow shows a report: which days were copied to, and per day *Exact copy* or *a of b trips as copied*, with the trips to plan by hand, each a link to it (*Show the trip on its day*).

The day follows a pattern

A day that was copied to **follows** the day it was copied from. You see that in a label on the day: *Follows Mon*, *Own changes* or *This date only*. What you can do:

Reset to pattern	Make the day as its pattern is again: its own changes go.
Copy it again	Its pattern changed since it was copied.
Stop following	This day keeps its planning as it is and follows no other day any more.

When copying doesn't work

MESSAGE	FIX
… run this planning already: nothing to copy there.	Nothing to do.
… has changes not saved	Save first: a copy takes the *saved* planning. Then try again.
Nothing of the planning matched on …	That day has another timetable. Copy the line's timetable there first (Timeplan › Copy to…), save and refresh.
Nothing was copied: a locked vehicle or duty there holds those trips.	Unlock it on that day and copy again.

MESSAGE	FIX
Not copied, their timetable is still being read	Wait a moment and try again.

✓ **THE ORDER FOR A WHOLE WEEK**

Finish Monday and save it with **Every day of this day type**: Tuesday to Friday are done. Then plan Saturday and save; then Sunday. Plan three days, run a whole week.

CHAPTER 11

Finding and fixing issues

RosterFlow checks your planning against the rules in Preferences all the time. Each issue is a question: fix it, or decide on purpose that it may stay.

Three kinds of issue

GLOBAL

The whole day

Trips without a bus, more buses than you have, more duties than drivers.

DUTY

One driver

Breaks, spread, length, changes of bus, where a duty begins or ends.

VEHICLE

One bus

Layovers, estimated deadheads, lines that may not interline.

The counts are at the bottom right of the status bar. A row with an issue has a red **!** before its ID: hover over it to read which rule is broken.

- HARD A hard rule. Optimise never breaks it; break it by hand and it is an error.
- SOFT A soft rule. Allowed, but every time it is broken a **penalty** is added to the cost. Optimise weighs the penalty against the cost.

Working with the issue list

1. **Open the issues**
   Click **Global Issues**, **Duty Issues** or **Vehicle Issues** in the status bar.
2. **Jump to an issue**
   Click an issue in the list: RosterFlow shows its row.
3. **Fix it**
   Move a trip, cut or join a duty, or change the rules under Preferences.
4. **Or dismiss it on purpose**
   Tick it and choose **Dismiss issues**. It then sits under **Dismissed issues**; **Restore issues** brings it back.

Filter issues shows only the rows with the issues ticked. **Select all** ticks everything at once. The round arrow in the status bar validates again.

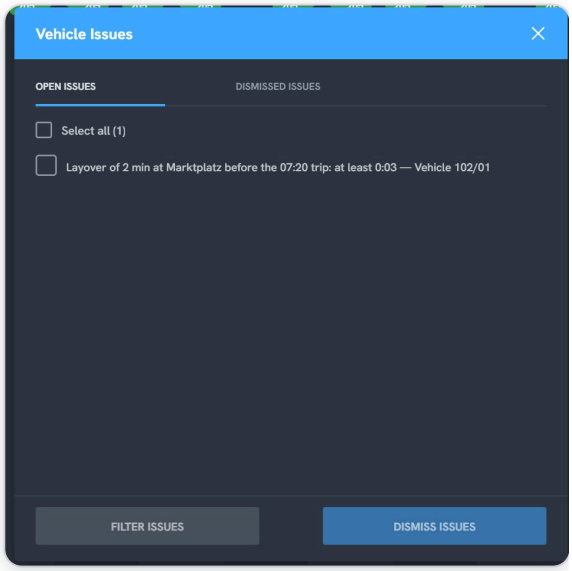


Figure 11.1 Vehicle Issues: a layover of 2 minutes at Marktplatz, shorter than the rule's 3 minutes.

Every issue, its cause and its fix

ISSUE	CAUSE	FIX
Trips not assigned	There are trips on the stack.	Give them a bus via the stack, or let Optimise fill the gaps.
More buses than the company has	The peak is higher than your fleet.	Buy or rent buses, spread trips out, or interline to lower the peak.
More buses than the depot holds	More buses out at once than Depot Setup allows.	Lower the peak, or raise the limit in Preferences.
More duties than drivers	The day has more duties than drivers on the staff.	Hire drivers on the Staff page, or make fewer, longer duties.
No meal break	A long duty has no break of the length needed.	Cut the work differently, or move a trip so a gap opens.
Meal break too late	The break only comes after more work than allowed.	Bring the break forward by swapping trips between duties.
Driving without a break	Too long a stretch of continuous work.	Make room for a break of the minimum length within the limit.
Spread over the limit	Sign-on to sign-off takes too long.	Cut the duty shorter, or make two duties of it.
Duties too short	Shorter than the least duty length.	Add work or join it with another duty. A duty too short is paid up to the least length.
Pieces too short	A piece on one bus is too short (costly).	Join the piece with a neighbouring piece on the same bus.
Too many changes of bus	The driver changes buses too often.	Keep them on one bus longer.
No time to get to the relief point	Duty travel takes longer than the gap.	Choose a later relief, or a faster way of travel.
Working time	A working-time rule of the company is broken.	Read the rule in the message and adjust the duty.
Duty type unsatisfied	The duty fits no allowed duty type (it is *Other*).	Change its start, end or length, or add a duty type.
Layovers too short	A bus turns faster than the least layover.	Move the next trip to another bus.
Deadheads estimated	RosterFlow doesn't know a deadhead's time and guesses it (orange dot).	Fill in the time under Preferences › Deadheads.
Duties begin or end at a wrong place	Signing on or off where that isn't allowed.	Allow it under Relief Points, or change the place in the duty's panel.
Breaks at a wrong place	A break where no break may be taken.	Allow breaks under Relief Points, or move the break.

ISSUE	CAUSE	FIX
Reliefs at a wrong place	A change of driver where no relief is allowed.	Join the duties, or allow relief at that place.
Buses on lines that do not interline	A bus runs trips of lines not ticked under Interlining.	Move the trips, or tick the line in Preferences › Interlining.

✓ **RED NUMBERS IN THE STATUS BAR?**
Then some trips have no bus or no driver. Look at the stack above the rows, or let Optimise fill the gaps. A red difference beside a KPI means your last change made it worse: `Ctrl` + `Z` turns it back.

CHAPTER 12

Preferences: your company's rules

Preferences are your company's labour agreement and your planning's laws of nature at once. The issues are checked against them, and Optimise works to them. Master the preferences, and you master RosterFlow.

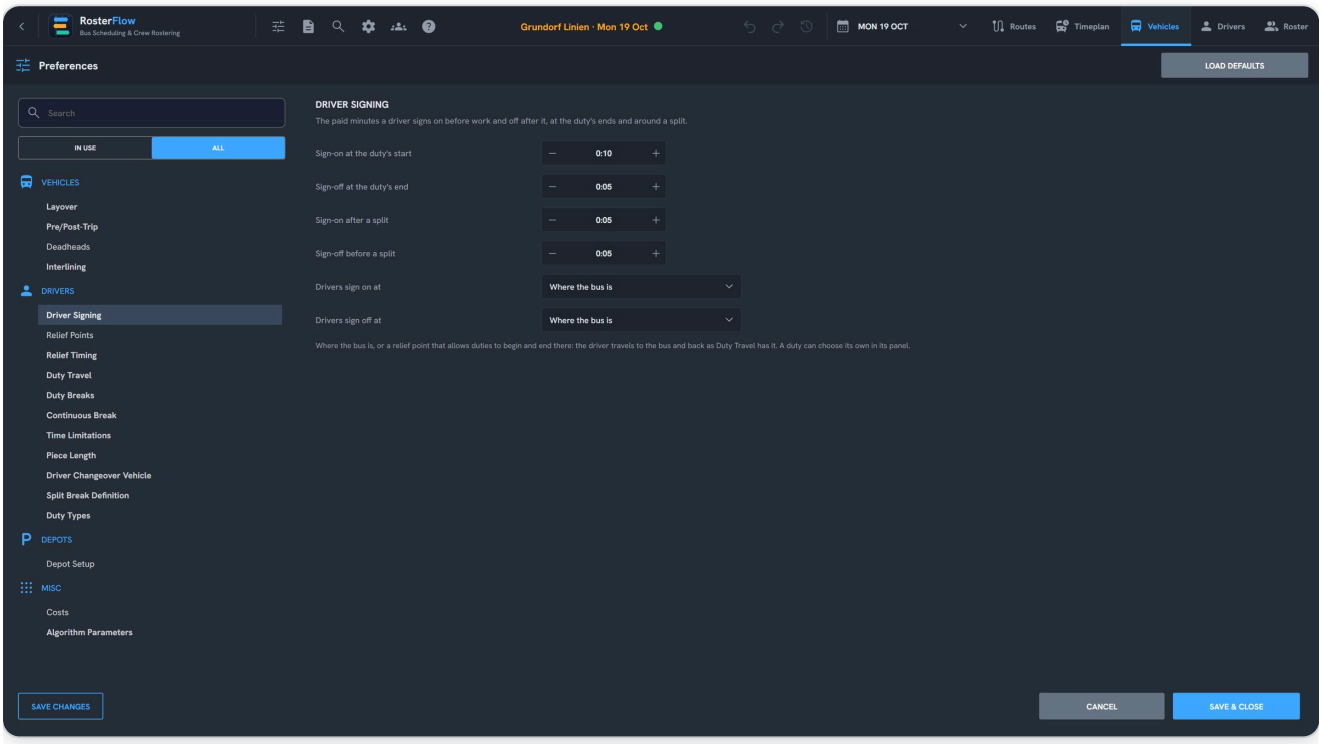


Figure 12.1 The Preferences window: the topics in four groups on the left, the chosen topic's settings on the right.

The window

- Open it with the first icon in the title bar (the sliders).
- On the left are the topics in four groups: **Vehicles**, **Drivers**, **Depots** and **Misc**. **Search** finds a topic by name; **In use** shows only the topics that are on, **All** shows everything.
- On the right the settings. Change values with **–** and **+**, or type them.
- At the foot of each topic a **live line**: *Nothing breaks this rule now* (green) or *3 duties break this rule now* (red), with in brackets the number with the saved rules. So you see the effect before you save.
- **Load defaults** sets everything back to the defaults. **Save changes** keeps them and leaves the window open; **Save & close** keeps and closes; **Cancel** forgets.

A rule: in use, hard or soft, penalty

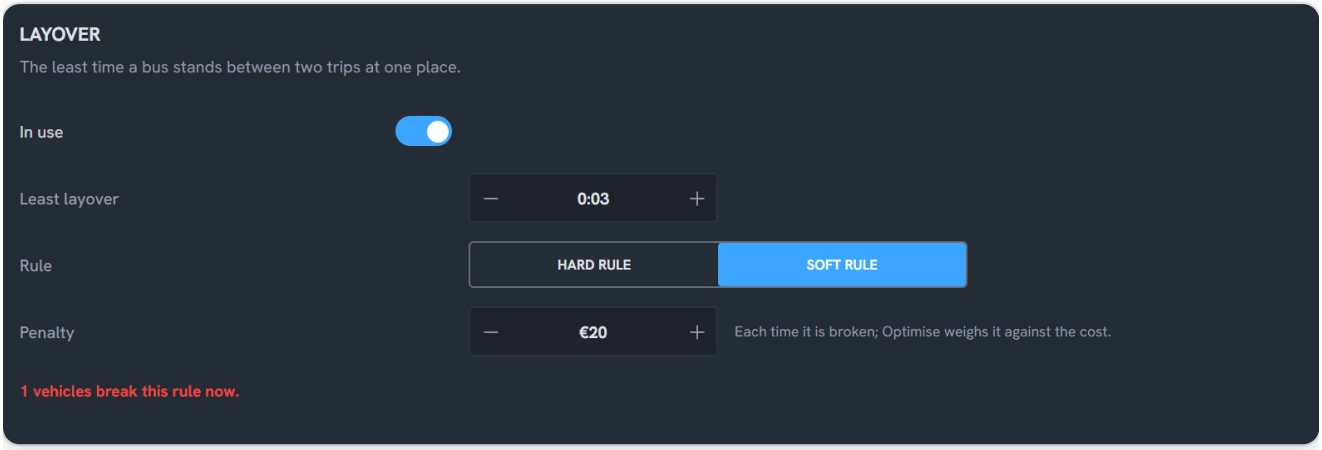


Figure 12.2 A rule with all its parts: in use, the value, hard or soft, and the penalty.

- **In use**: switches the rule on or off.
- **Rule**: a `HARD RULE` is never broken by Optimise and is an error when broken by hand. A `SOFT RULE` may be broken, at a penalty.
- **Penalty**: what it costs, each time the soft rule is broken. Optimise weighs it against the cost.

> **AFTER SAVING**
> If the changed rules break something in the planning, RosterFlow says so at once: *Broken now by the changed rules: …*, with the button **Optimise them**.

Vehicles

Layover

The least time a bus stands between two trips at one place. Default **0:03**, soft rule, penalty €20. Too short a layover makes your service unreliable: one minute of delay carries on all day.

Pre/Post-Trip

The bus's check before it leaves the depot (**Pre-trip check**, 0:05) and after it is back (**Post-trip check**, 0:05).

Deadheads

How long a bus takes to drive empty from one place to another, and how far. A default (0:05 and 3 km) for where nothing is set; below that per relief point **to or from it**, and a table **between two relief points** (minutes or kilometres).

An empty field takes the way back's value, else the relief points' own, else the line's fastest trip between the two, else the default. A deadhead whose time is set here is no longer estimated: no orange dot.

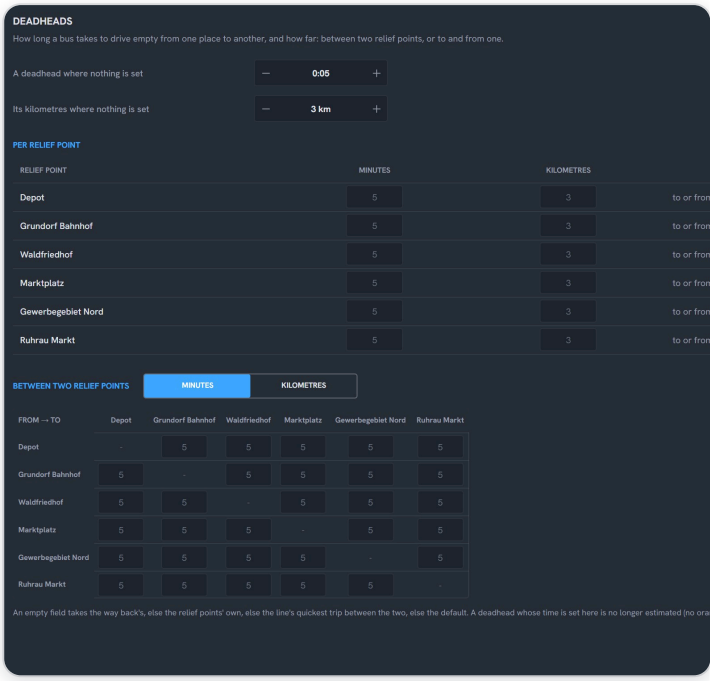


Figure 12.3 Deadheads: the default and per relief point.

Interlining

A bus may run trips of several lines in its day. With **Buses interline** on, Optimise pairs a trip of one line with one of another where the one ends and the next begins, or with a short deadhead. **Longest run empty between two lines** (0:15 by default) limits that deadhead. Under **Lines that interline** you tick per line whether it may interline; a line not ticked keeps to its own buses. Optimise makes no deadhead whose time it doesn't know.

Drivers

Driver Signing

The paid minutes around the work: **sign-on at the duty's start** (0:10), **sign-off at the duty's end** (0:05), and around a split (0:05 and 0:05). Plus **where** drivers sign on and off: *Where the bus is*, or a relief point where duties may begin and end. The driver then travels to the bus and back as Duty Travel says. A duty can choose its own place in its panel.

Relief Points

Where drivers may begin or end a duty, change on the bus and take a break: the depot and the lines' termini by themselves, and stops on the way that you add. Three ticks per point: **Sign-on/off**, **Relief** and **Break**.

With **Add a stop on the way** you choose a line and a stop and add it. After that, drivers can change at that stop in the middle of a trip.

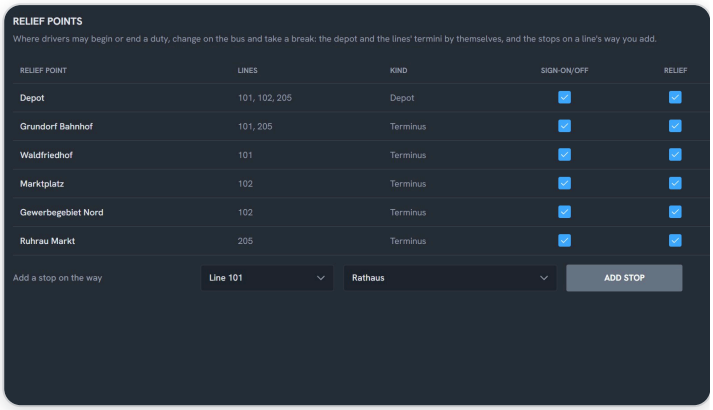


Figure 12.4 Relief points with their three permissions.

Relief Timing

When the driver changes within the bus's layover at a relief point: **on arrival and on departure (unattended)**, **on arrival**, **on departure**, **mid-layover** or **before departure**.

Duty Travel

How a driver gets to a relief point at another place, and how long it takes. The **way by default** (walk, public transport or staff car) and the minutes per way: **Walk (W)** 0:05, **Public transport (P)** 0:15, **Staff car (R)** 0:10. Below that, a way and minutes of its own per relief point, and a table between two relief points.

Duty Breaks

Show a gap as a break from (0:15): shorter gaps are standing time. **A duty working longer than** 6:00 **needs a meal break of** 0:30: a hard rule by default. The meal break must begin before the duty has worked that long.

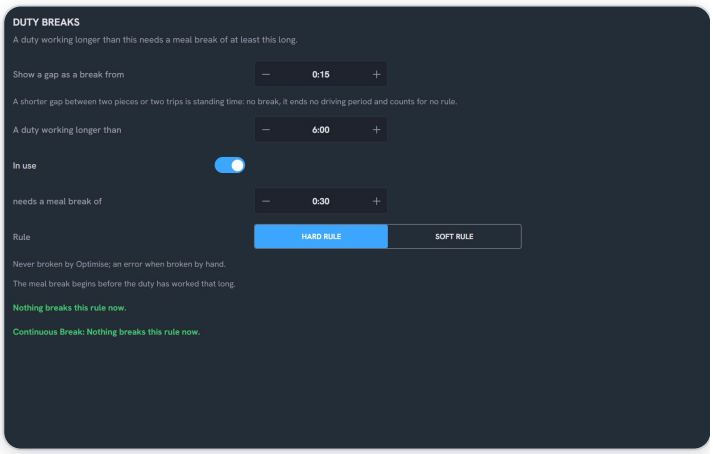


Figure 12.5 Duty Breaks.

Continuous Break

The longest a driver works without a break of at least a given length: **Longest work without a break** 4:30, hard rule; **A break is at least** 0:15. With **A layover with the bus counts as a break** on, waiting at the terminus counts too.

Time Limitations

Spread at most 13:00 (soft rule, €50) and **Duty length at least** (off by default). A shorter duty is an issue and is paid up to the least length.

Piece Length

Pieces at least 1:00, soft rule, €10: short pieces are costly, because every piece costs signing or travel time.

Driver Changeover Vehicle

Changes of bus at most 2 per duty, soft rule, €15.

Split Break Definition

A split is a break of at least 1:00. A break between two pieces this long or longer is a split: unpaid, the driver signs off and on again.

Duty Types

The kinds of duty the company works, such as **Early shift**, **Middle shift**, **Late shift**, **Split shift** and **Night shift**. Per type:

- **Allowed** on or off;
- **Start time**: the window in which the duty starts (for example 03:00–07:59);
- **End by**, **Length at most**, **Length at least**;
- **Split**: no, yes or either;
- **Penalty**.

The **first** type in the order that a duty fits is its type (move types with the arrow). If a duty fits no allowed type it is **Other**: an issue. **New type** adds one.

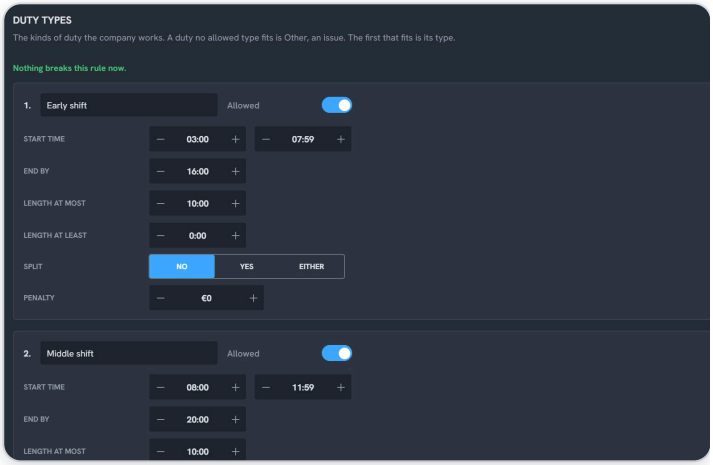


Figure 12.6 Duty Types: the early shift.

Depots and misc

Depot Setup

Buses out at once at most: how many buses the depot holds. Off by default.

Costs

What the crew and the vehicles cost; Optimise works the cost down. By default RosterFlow takes the amounts **as the company pays**. Tick **Set** to use an amount of your own for: **a driver's hour**, **each duty besides its hours**, **each bus out a day** and **each kilometre**.



Figure 12.7 Costs, as the company pays.

> **TIP: STEER OPTIMISE WITH COSTS**
> Make a bus per day expensive, and Optimise will save buses, even if that costs more driver hours. Make a duty expensive, and it prefers fewer, longer duties. That way you steer the result without moving a single trip by hand.

Algorithm Parameters

What Optimise makes: **Optimise vehicles (the blocks)** and **Optimise duties (cut and paired)**. The same as the gear beside the button.

CHAPTER 13

Keeping the overview

A real day has dozens of buses and duties. With stats, layers, filters and Go to you keep the overview.

Stats

Adds columns of figures beside the rows: in Vehicles for example **Start time**, **End time**, **Spread**, **Driving time**, **Distance** and **Trips**; in Drivers figures such as **Paid time** and **Duty type**. Click a column heading to sort by it, again the other way, a third time back. **Reset** takes the columns away.

> **TIP**
> Sort duties by paid time to find the outliers, or vehicles by distance to see the bus with the most kilometres.

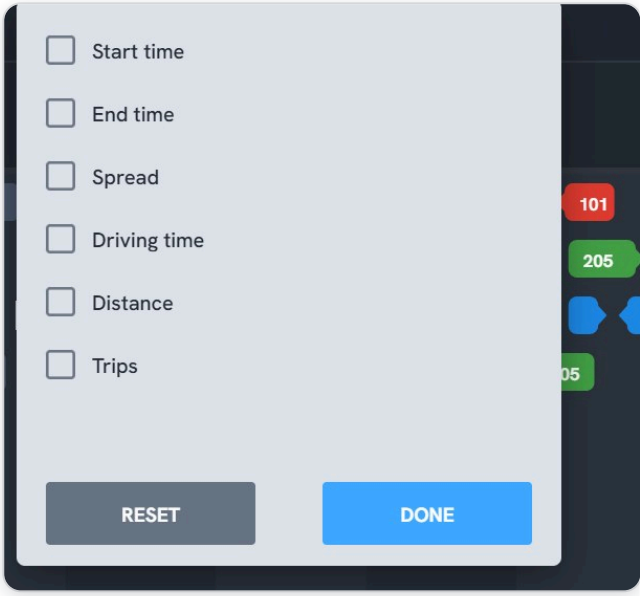


Figure 13.1 Stats in Vehicles.

Layers

Layers chooses what the chips say and how they are coloured.

- **Colour by**: *Line (route)*, *Trip ID*, *Block (vehicle)* or *Duty*. **Show the legend** tells you what each colour is. Colouring by duty in Vehicles shows at a glance where drivers change.
- **Trips**: what a chip says: the *route*, the *start and end time* or the *start and end stops*.
- **Between trips**: show the *break duration*, and *mark trips off their timetable tour* (a dot on a trip moved away from its timetable block).
- **Duties**: *Driving periods under the duties*.

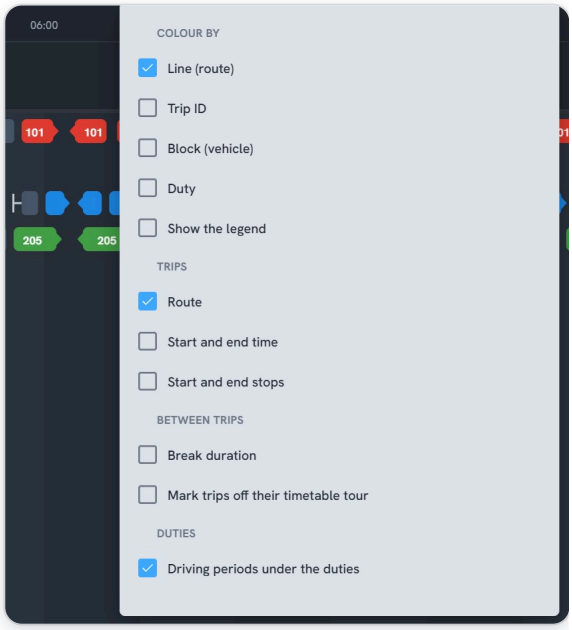


Figure 13.2 Layers.

Filter

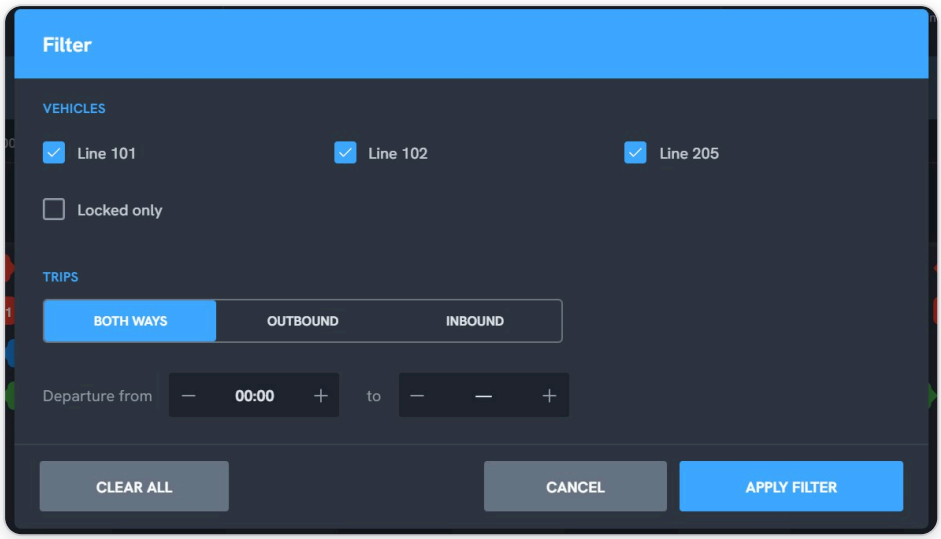


Figure 13.3 Filter: lines, locked only, direction and departure time.

Show only certain **lines**, **locked only** rows, one **direction** (both ways, outbound or inbound) or the trips with a **departure from … to ….** **Apply filter** switches it on, **Clear all** takes it away. From the issue list you can also filter by issues.

Go to, pinning, full screen

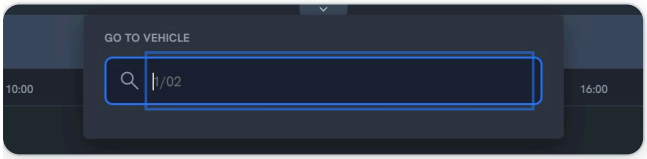


Figure 13.4 Go to vehicle or duty (G).

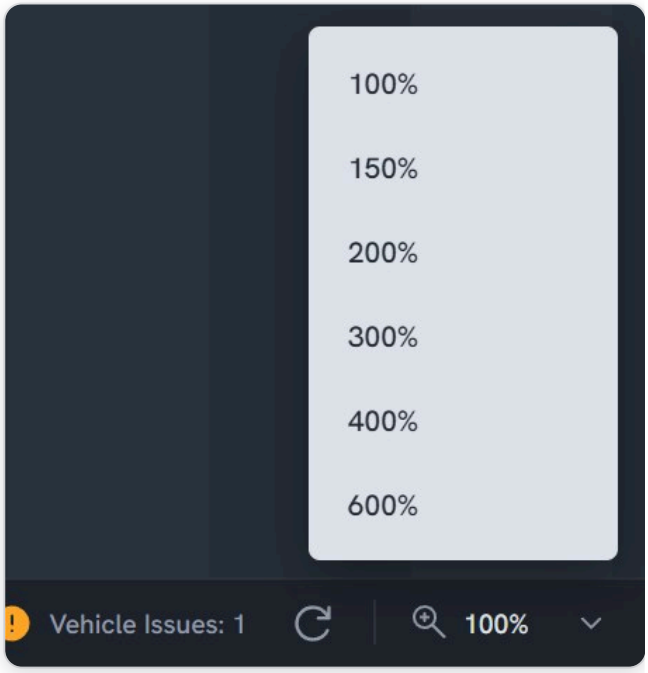


Figure 13.5 The zoom: 100 to 600%.

- G or the magnifier: type a vehicle's or duty's ID and press Enter.
- P or the pin: pin the chosen row to the top; click again to unpin.
- The corners on the right of the toolbar: **full screen**, with the KPIs folded away.

CHAPTER 14

Printing

As at a real operator, you can print the day as a graph: for yourself, to share with fellow players or to pin on the wall.

Click the document icon in the title bar and choose:

- **Bus graph (PDF)**: every vehicle's day on the hours, the way professional scheduling software prints it.
- **Duty graph (PDF)**: every driver's duty on the hours.

Each graph has a header with the **depot**, the **service** (day type), the **date** and when it was **printed**. RosterFlow asks where to save the file and says *Saved: …*, with **Open the file**.

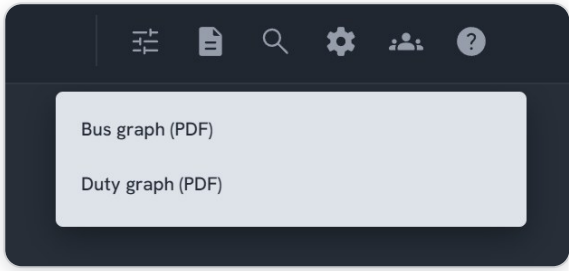


Figure 14.1 Print: bus or duty graph.

COMPUTER ONLY
Printing works in the app on your computer, not in the browser on omsihub.com.

TIP: PLANNING ON PAPER
Print the duty graph before you optimise, and once more afterwards. Side by side you see exactly what changed.

CHAPTER 15

RosterFlow in the game and planning together

What you save, the game runs. This chapter explains how your planning comes to life on the company day, what happens when something is missing, and how to plan together with others.

The company day runs the saved planning

- The roster, the company day and the buses on the map follow the **saved** blocks and duties.
- Duties without a driver in the roster: *their trips are dropped until they get one* (the Roster tab, Fill the roster). Likewise a tour without a bus.
- If RosterFlow had unsaved changes for the day, the company says **RosterFlow not saved**: the day ran the saved planning. Save or discard, so the day runs what RosterFlow shows.
- Trips the day didn't run are listed with the reason, such as *The bus broke down on the way*, *The driver came late* or *On no bus: the trips wait on RosterFlow's stack*. Click one to see it in the planning.

Driving yourself

Plan yourself (**You**) on a duty in the Roster to drive it yourself. The duty card and the company's launcher then open exactly that duty. If the trips no longer match (the timetable or planning changed since), the game asks you to choose the duty again in RosterFlow. If the tour has no bus from the fleet, you give it one in RosterFlow.

Planning together

In a shared company (multiplayer) RosterFlow works fully together: what one does, the other sees at once.

- When someone else changes something, you see for example *Anna changed the blocks and duties of Mon 19 Oct* or *… changed the preferences*.
- Your Undo then begins again from that point: you can't undo someone else's changes.
- If someone changed the day while you were doing something, it is shown anew: do it once more.
- Your **role** decides what you may do. With a role that may only look, the title bar says **Read-only**; the boss can give you *Edit the planning* on their Multiplayer page.
- Routes are drawn on the boss's map, with their line editor. Lessons only work on the boss's own launcher; the guided tour and the help sheet work for everyone.
- The people icon in the title bar opens **Play together**: friends drive duties of your company in your world. They take open duties, or you give them one in the Roster.

CHAPTER 16

Learning in the studio

Behind the question mark in the title bar is the built-in tutorial: a guided tour of every part of the screen, eight lessons to try yourself and a one-page help sheet.

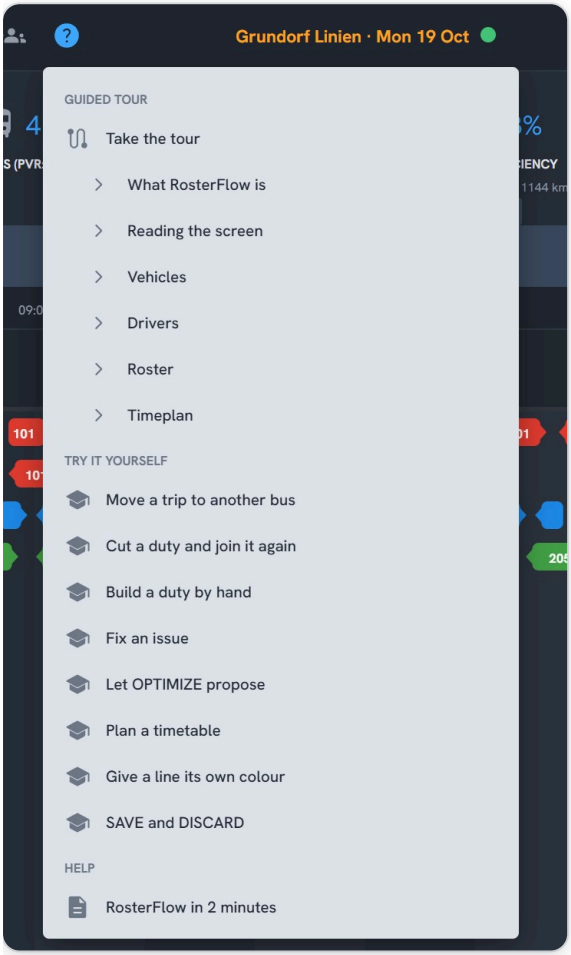


Figure 16.1 The tutorial's menu.

The guided tour

Take the tour starts from the beginning; or choose a chapter: *What RosterFlow is*, *Reading the screen*, *Vehicles*, *Drivers*, *Roster* or *Timeplan*. Each step highlights a part of the screen. → or **Enter** goes on, ← back; **Skip** stops.

Try it yourself: the lessons

1. Move a trip to another bus
2. Cut a duty and join it again
3. Build a duty by hand
4. Fix an issue
5. Let OPTIMIZE propose
6. Plan a timetable
7. Give a line its own colour
8. SAVE and DISCARD

A lesson shows a card with its steps and ticks them off while you work. At the end you choose **Keep my changes** or **Undo the lesson**: then the planning is as it was before.

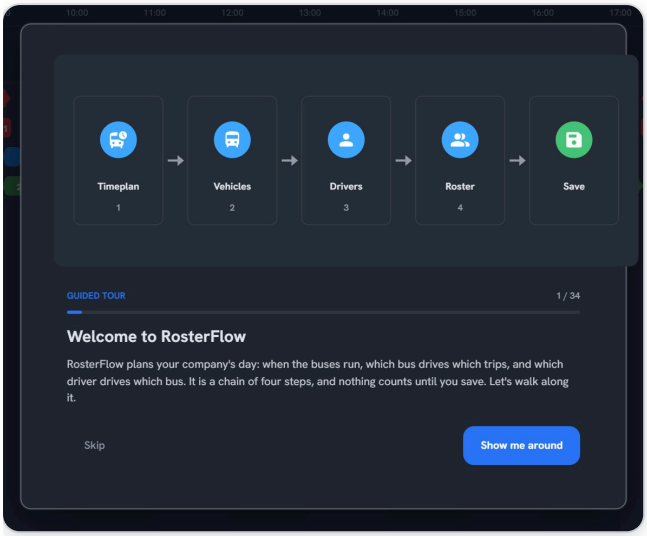


Figure 16.2 The tour's first step: the chain.

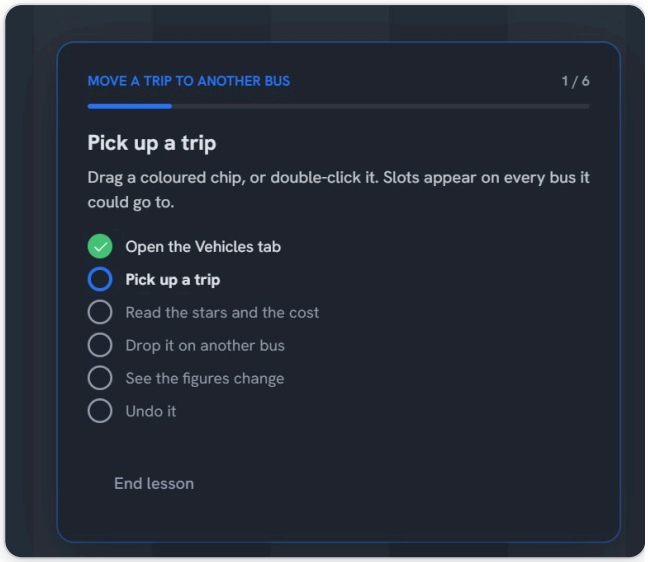


Figure 16.3 A lesson: every step is ticked off.

The help sheet

RosterFlow in 2 minutes: the chain, the terms with their symbols, the shortcuts and what to do when something is red, on one page. Ideal to keep open beside your planning.

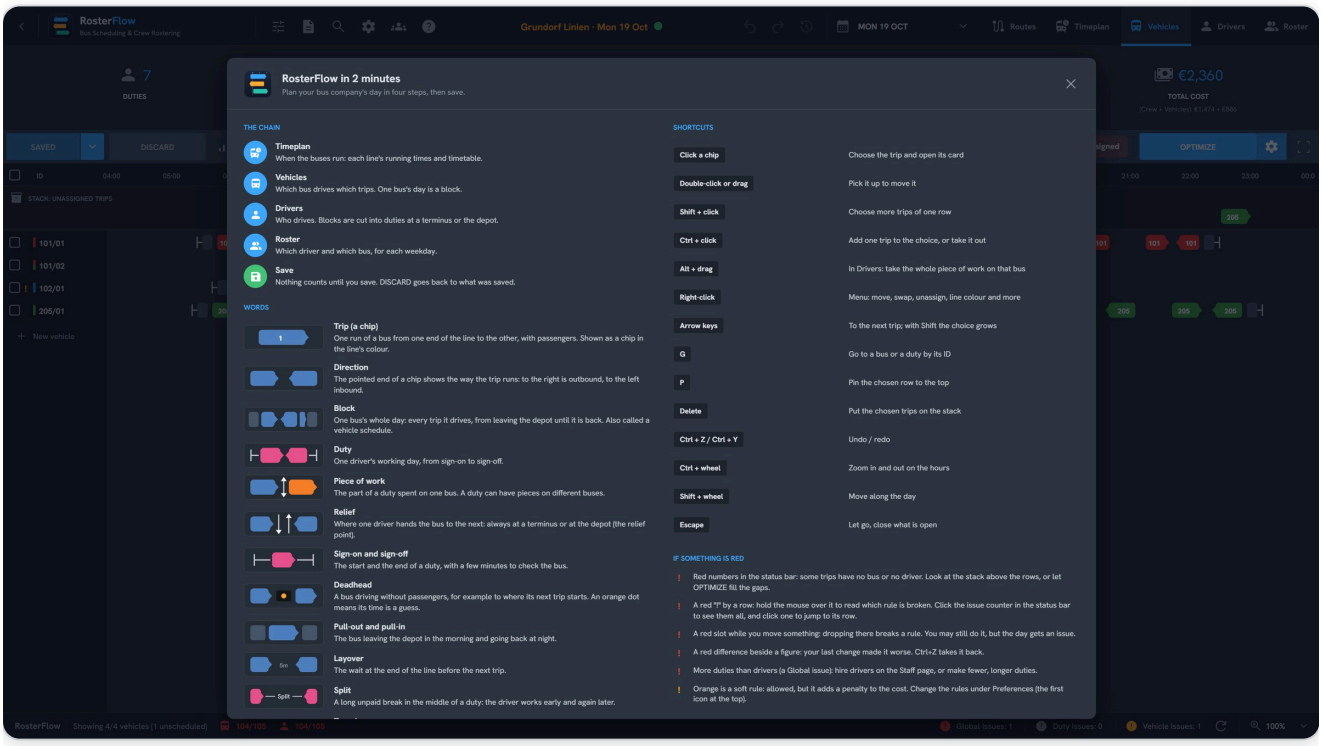


Figure 16.4 The help sheet “RosterFlow in 2 minutes”.

04

PART IV

Apply

Complete workflows for situations you meet as a scheduler, exercises to test your skill, answers to frequently asked questions and all the reference in one place.

17 Workflows

18 Exercises

19 Frequently asked questions

A-C Appendices: shortcuts, defaults, checklist

CHAPTER 17

Workflows

Recipes from start to finish. Each one names the goal, the steps in order and what to watch. They use everything from the earlier chapters.

Workflow 1 · From new line to a running service

Goal: a new line that runs with buses and drivers tomorrow.

1. **Routes**
   Make the route: stops out and back, days and colour. Confirm and pay.
2. **Timeplan › Running times**
   Check the minutes from the roads. Make time bands for the peaks (for example 07:00 and 16:00) and add a minute or two there.
3. **Timeplan › Timetable**
   Add trips by headway, per direction: the peaks first, then the off-peak and the evening. Check the issues button at the bottom right: zero.
4. **Copy to other day groups**
   With **Copy to…** to Saturday and Sunday (and adjust the headway there), or make them separately.
5. **Save and Refresh**
   Save the timetable and click **UNSYNCED – Refresh**.
6. **Vehicles**
   Check the blocks. If needed, let OPTIMIZE make a proposal, and accept it if the figures are better.
7. **Drivers**
   Check the duties for red exclamation marks. Fix issues or let Optimise make the duties.
8. **Save on the week**
   Arrow beside Save › **Every day of this day type**. Repeat for Saturday and Sunday.
9. **Roster**
   Fill the roster for every day, and check the open duties on the right. Hire drivers if there are too few.

✓ RULE OF THUMB FOR LAYOVERS

At every terminus, plan at least the least layover plus a small buffer. A bus that always has a few minutes to turn catches up on delays; a bus without a buffer carries them all day.

Workflow 2 · Saving a bus

Goal: run the same trips with one bus fewer.

1. **Look at the peak**
   The number in brackets by Blocks is the peak. That is what must come down.

2. **Find the short blocks**
   Sort with Stats by **Trips** or **Driving time**. A block with few trips is a candidate to remove.
3. **Share out its trips**
   Pick up its trips and put them on other buses, at the best stars. Use **Swap** or **Swap from here** to make room.
4. **Allow interlining**
   If it doesn't work within one line, switch on lines that end at the same place in Preferences › Interlining.
5. **Or let it calculate**
   Switch on only **Optimise vehicles** in the gear and click Optimize. If you like, raise the cost per bus per day to make buses weigh extra heavily.
6. **Check the duties**
   Fewer buses change the duties. Check in Drivers that everything still fits, and save.

Workflow 3 · Fair duties with good breaks

Goal: no duty without a meal break, no working days that are too long.

1. **Set the rules**
   Preferences › Duty Breaks, Continuous Break and Time Limitations. Make what really must hold a hard rule.
2. **Decide your duty types**
   Early, middle, late, perhaps split. Set types you don't want to not allowed.
3. **Choose the relief points**
   The more places drivers may change, the more freedom to place breaks well. Add a stop on the way if needed.
4. **Optimise the duties**
   Switch on only **Optimise duties** in the gear.
5. **Read the driving periods**
   Switch on *Driving periods under the duties* in Layers and hover over the bands: all within the limit?
6. **Lock and save**
   Lock good duties, refine the rest by hand, save.

Workflow 4 · A holiday or event

Goal: let one date run differently without changing the usual week.

1. **Choose the day**
   Choose that date in the day switcher.
2. **Adjust the planning**
   For example an extra bus for an event: drop trips on **+ New vehicle**.
3. **Save for this date**
   Arrow beside Save › at the top **Only [date]**. The weekday keeps its usual planning.

4. **Another timetable?**
   If the day runs another timetable (such as Sunday's), the Timeplan's day strip shows it: *Runs the Sun timetable, not its weekday's: a holiday or another day*.

Workflow 5 · A driver calls in sick

Goal: lose as few trips as possible today.

1. **Roster, company day**
   Open the Roster tab on the company day. Take the driver off (cross or `Delete`) if they haven't started yet.
2. **Give the duty to someone else**
   Click the open duty; the list on the right shows who could take it. Give it to a free driver.
3. **No free driver?**
   Right-click › **Give to an agency driver**, or drive it yourself (**You**).
4. **Prevent it next time**
   Plan a spare line in the peak in Drivers.

Workflow 6 · The week in three days

Goal: a complete week with as little work as possible.

1. **Monday**
   Plan Monday fully: blocks, duties, no issues.
2. **Copy to the day type**
   Save › **Every day of this day type**: Tuesday to Friday follow Monday.
3. **Saturday and Sunday**
   Plan them separately (another timetable), or copy and plan the listed trips by hand.
4. **Roster**
   Fill Monday's roster and choose **Repeat** to copy it to the other weekdays.
5. **Changes later**
   If you change Monday, save again with the day type, or choose **Copy it again** on a day that follows.

CHAPTER 18

Exercises

You learn by doing. Each exercise has a task and a checkpoint: that way you know you did it right. Discard afterwards, and nothing can go wrong.

PRACTISING SAFELY

Practise freely in Vehicles and Drivers: as long as you don't save, the game runs the saved planning. Done? Click DISCARD and everything is back. In the Timeplan, only save when you mean it.

Level 1 · Basics

EXERCISE 1 · FOLLOWING THE CHAIN

Open RosterFlow and visit the five tabs from left to right. For each tab, say out loud what it plans.
Check: you can explain what a block and a duty are without looking.

EXERCISE 2 · READING THE KPIS

Fold out the extended figures. How many split duties does your day have? What is the average paid time?
Check: you find both numbers again in the Duties paid time histogram.

EXERCISE 3 · MOVING A TRIP AND TURNING IT BACK

In Vehicles, move a trip to a slot with five stars. Note the cost difference. Undo it with Ctrl + Z.
Check: the differences beside the KPIs are gone and the dot is green again.

EXERCISE 4 · THE STACK

Unassign a trip (Delete). Look at the status bar and the issues. Give the trip another bus via the stack.
Check: the bus number in the status bar equals the total again.

Level 2 · Intermediate

EXERCISE 5 · CUTTING AND JOINING

In Vehicles, cut a block at a terminus. Look at the new duty in Drivers. Join the duties again with the arrows.
Check: the number of duties is one higher first, then the same again.

EXERCISE 6 · BUILDING A DUTY

In Drivers, use Alt +drag to make a new duty from two pieces of different duties, with a break in between.
Check: the new duty has no red "!" and its panel shows two pieces of work and a break.

EXERCISE 7 · FIXING AN ISSUE

Open the issues, jump to one and fix it without dismissing it.

Check: the count in the status bar is one lower.

EXERCISE 8 · PEAK RUNNING TIMES

In the Timeplan, make a 07:00–09:00 time band with one minute more on every stretch. Look at the trips in the timetable. Undo it.

Check: that band's trip time was higher, and all its trips arrived later.

Level 3 · Expert

EXERCISE 9 · BEATING THE OPTIMISER

Click Optimize and note the proposal's cost. Keep your own planning and try to get closer to that cost by hand, with Swap and Swap from here.

Check: you are within 5% of the proposal. Better? Then your knowledge of the day found something the rules don't say.

EXERCISE 10 · STEERING WITH COSTS

In Preferences › Costs, raise the amount for each bus out a day sharply, and optimise. Compare the number of blocks and duties with the previous proposal. Set the amount back afterwards.

Check: you can explain why the number of buses fell and the driver hours rose.

EXERCISE 11 · A WEEK IN THREE DAYS

Follow workflow 6 and read the report after copying. Plan the trips to plan by hand.

Check: after Fill the roster no open duties are left on the right of the roster, on any day, and the status bar has no red numbers.

EXERCISE 12 · THE BUILT-IN LESSONS

Do all eight lessons in the question-mark menu, choosing **Undo the lesson** each time.

Check: every lesson ends with *Well done!*

CHAPTER 19

Frequently asked questions

The questions players ask most often, with a short answer and where to read more.

I planned everything, but the game runs something else.
You didn't save: the dot is orange. Click SAVE. See chapter 10.

I changed the timetable, but the buses run the old times.
After saving in the Timeplan you still have to **Refresh** (UNSYNCED in the status bar). Then check and save the blocks and duties. See chapter 5.

My new line doesn't run.
A line runs no tour until it has a timetable. Make one in the Timeplan, save and refresh. Then check in the Roster that the tour has a bus and the duties a driver.

I can't drop a trip anywhere: no slot appears.
The trip fits no bus in time. Make room by swapping a trip, or drop it on **+ New vehicle**.

In Drivers there is no slot by the duty I want.
That duty would break a rule (usually too long), or the drivers can't change there. Take a shorter piece, or add a relief point. See chapter 7.

I can't cut anywhere: no blue line.
A cut can only be made at a relief point that allows relief. Check Preferences › Relief Points.

Optimise says there is no better plan, but I think the planning is bad.
Optimise measures “better” by your Preferences. Change the costs or rules so they say what matters to you (chapter 12), and optimise again.

Optimise changed a duty I made by hand.
Lock rows that must stay. A proposal is only taken over with Accept; Keep mine forgets it.

The status bar says 104/105 in red.
One trip has no bus or no driver. Look on the stack, or let Optimise fill the gaps.

My deadheads have an orange dot.
RosterFlow guesses their time. Fill in the minutes under Preferences › Deadheads, and they are no longer estimated.

Copying to Saturday took only half the trips.
Saturday runs another timetable. Only trips with the same line, way, departure and stops match. Plan the rest by hand, or copy the timetable to Saturday first.

I want something only for next Tuesday.
Save with **This date only** in the Save and apply to other days window.

The Roster doesn't show my new duties.
The Roster works with saved duties. Save in Vehicles or Drivers first.

I can't change anything in multiplayer.
Your role is read-only. Ask the boss for *Edit the planning*.

I'm lost in a big day.
Use G (Go to), the Filter by line, Layers › Colour by duty, and pin your most important rows with P.

APPENDIX

A · Keyboard shortcuts

Every key of RosterFlow on one page.

Vehicles and Drivers

KEY	WHAT IT DOES
Click a chip	Choose the trip and open its card
Double-click or drag	Pick it up to move it
Shift **+ click**	Choose more trips of one row
Ctrl **+ click**	Add one trip to the choice, or take it out
Alt **+ drag**	In Drivers: take the whole piece of work on that bus
Right-click	Menu: move, swap, unassign, line colour and more
← →	To the trip before or after; with Shift the choice grows
↑ ↓	To the row above or below
G	Go to a bus or duty by its ID
P	Pin the chosen row to the top
Delete	Unassign the chosen trips to the stack
Ctrl + C / Ctrl + V	Copy ticked rows, and paste them on another day
Ctrl + Z / Ctrl + Y	Undo / redo (also Ctrl + Shift + Z)
Ctrl **+ mouse wheel**	Zoom the hours in and out
Shift **+ mouse wheel**	Move along the day
Esc	Let go, close what is open; with nothing open: back to the company

Timeplan

Enter	Keep the cell
Tab	To the next cell
↑ ↓ **in a cell**	One minute more or less
Esc	Leave the cell without keeping; clear the choice
Click a trip number	Choose the trip; Shift +click a range, Ctrl +click one more or less

Ctrl + A	Choose every trip of the day
Delete	Take the chosen trips out
Ctrl + Z / Ctrl + Y	Undo / redo

Roster and tutorial

Delete	Roster: take the chosen drivers and buses off
Ctrl + Z / Ctrl + Y	Roster: undo / redo
→ or Enter / ←	Guided tour: next / previous step

APPENDIX

B · Every preference and its default

Every rule in Preferences with its default value, as you see them after **Load defaults**. The values come from the sample company; depending on your game mode and company some may differ.

TOPIC	SETTING	DEFAULT
Layover	Least layover	0:03 · SOFT €20
Pre/Post-Trip	Pre-trip / post-trip check	0:05 / 0:05
Deadheads	Where nothing is set	0:05 · 3 km
Interlining	Buses interline · longest run empty	on · 0:15 · all lines
Driver Signing	Start / end / after and before a split	0:10 / 0:05 / 0:05 / 0:05
	Sign on at / sign off at	Where the bus is
Relief Points	Depot and termini	sign-on/off, relief and break allowed
Relief Timing	When drivers change	On arrival and on departure (unattended)
Duty Travel	Way by default · W / P / R	Walk · 0:05 / 0:15 / 0:10
Duty Breaks	Show a gap as a break from	0:15
	Meal break after · length	6:00 · 0:30 · HARD
Continuous Break	Longest work without a break · break at least	4:30 · 0:15 · HARD
	Layover counts as a break	on
Time Limitations	Spread at most	13:00 · SOFT €50
	Duty length at least	off
Piece Length	Pieces at least	1:00 · SOFT €10
Driver Changeover Vehicle	Changes of bus at most	2 · SOFT €15
Split Break Definition	A split from	1:00
Duty Types	Early, middle, late, split, night	as the company has them
Depot Setup	Buses out at once	off
Costs	Hour, duty, bus a day, kilometre	as the company pays
Algorithm Parameters	Optimise vehicles / duties	both on

APPENDIX

C · Checklist for a good planning

Run through this list before you let the day run. Every tick is one issue fewer on the company day.

TIMETABLE

- [ ] Every line has a timetable for every day group it runs on
- [ ] The issues button at the bottom right is at zero
- [ ] The timetable is saved
- [ ] No UNSYNCED left: refreshed

VEHICLES

- [ ] The stack is empty
- [ ] The peak is no higher than your fleet
- [ ] No layovers under the limit
- [ ] No estimated deadheads (orange dots)

DRIVERS

- [ ] No red “!” by any duty
- [ ] Every long duty has a meal break
- [ ] No more duties than drivers
- [ ] A spare line in the peak

SAVING AND ROSTER

- [ ] The dot in the title bar is green
- [ ] Saved for every day of the day type
- [ ] Roster filled for the whole week
- [ ] No open duties on the right of the roster

✓ **FINALLY**

A planning is never finished. After a few played days, look at which trips were dropped and why, and refine. Every round your day gets cheaper, more reliable and fairer to your drivers. Enjoy your planning!